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Azure Virtual Network troubleshooting documentation

Welcome to Azure Virtual Network troubleshooting. These articles explain how to determine, diagnose, and fix various issues that you might encounter when you use Azure Virtual Network. In the navigation pane on the left, browse through the article list or use the search box to find issues and solutions.

Azure Virtual Network

HOW-TO GUIDE

Virtual Network troubleshooter

[SDP3][fdcb6045-5616-45b4-bb68-0bd11081c184] Microsoft Azure Virtual Network Troubleshooter package

Article • 04/12/2024

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Original KB number: 2996010

Summary

This article describes how to troubleshoot issues with Microsoft Azure virtual networks by using a troubleshooter package. This diagnostic package takes advantage of the capabilities of the Azure PowerShell module and Azure Virtual Network representational state transfer (REST) APIs to perform the following tasks:

- Identify both the Azure virtual network (VNET) and the local network experiencing the VPN issue - List only Azure VNETs that have gateways, and we show only local network connections that are specific to the selected Azure VNET. If the customer selects a VNET that has no local network connection, or if the customer has no VNETS that have gateways, the diagnostic provides the customer with links to relevant documentation.
- Network configuration XML parsing - Parses the network configuration XML and includes it in a report in order to make it easier to understand and summarize customers' configurations.
- Azure Gateway Diagnostics - Similar to the tool we recently released for public download, the diagnostic package creates a three-minute Azure Gateway diagnostic file and asks the customer to turn on logging on the on-premises device and to reproduce the issue that he or she is experiencing.
- On-premises device configuration file and logging file - Prompts the customer to provide us with the device configuration and log file (after we ask the customer to reproduce the VPN issue).
- Virtual Network features configuration data collection :
  - Network Security Group (NSG)
  - Static IP (VNet and public)
  - Instance-level public IP (PIP)
  - ExpressRouteDNS (VNet, instance, and subnet level)

Package information

A supported package is now available from Microsoft Support. To obtain the Microsoft Azure Virtual Network Troubleshooter package, submit a request to Microsoft Online Customer Services. To do this, go to the following Microsoft Support website:

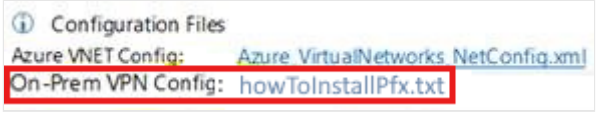
https://go.microsoft.com/?linkid=6294451

> **Note**
>
> If additional issues occur, or if any troubleshooting is required, you might have to create a separate service request. The usual support costs will apply to additional support questions and to issues that do not qualify for this specific package. To create a separate service request, go to the following Microsoft website:
>
> https://support.microsoft.com/contactus/?ws=support

More information

Information that is collected

- Network configuration XML
- On-premises VPN device configuration



- On-premises VPN device log

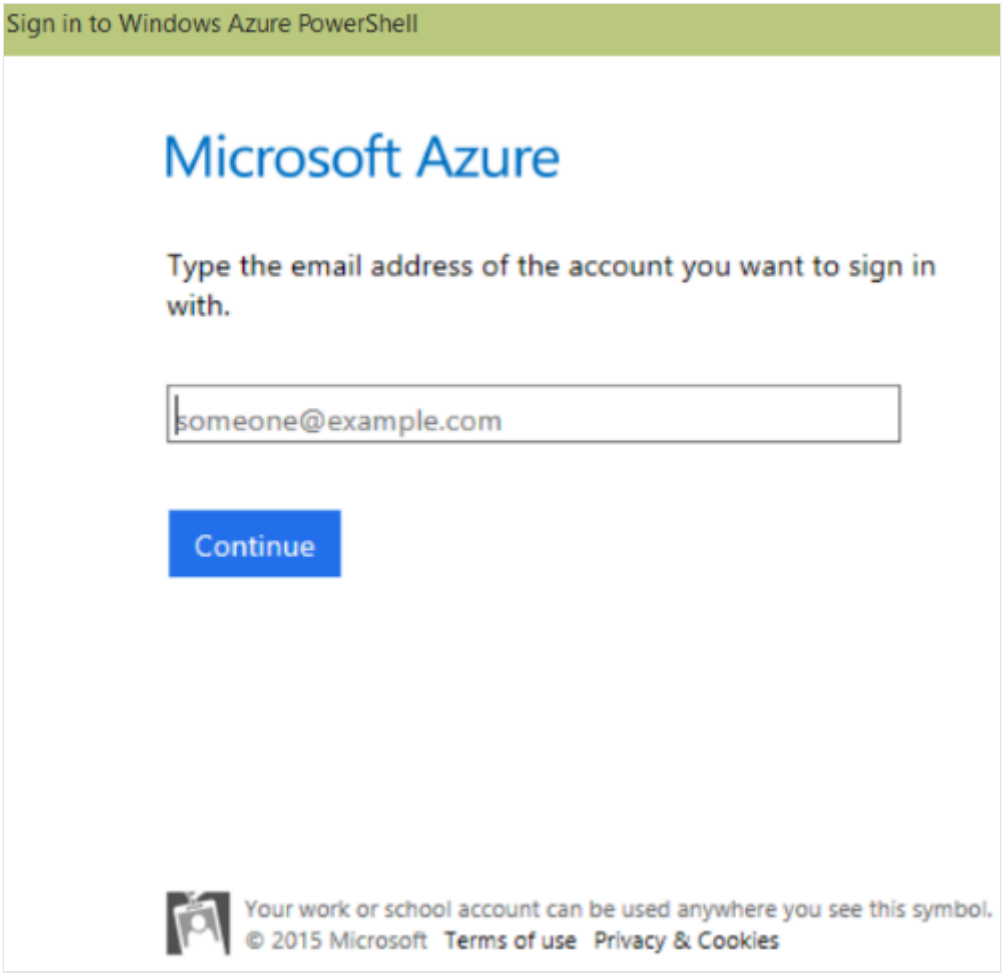
- Azure Gateway diagnostics log

- Azure Gateway connection statistics

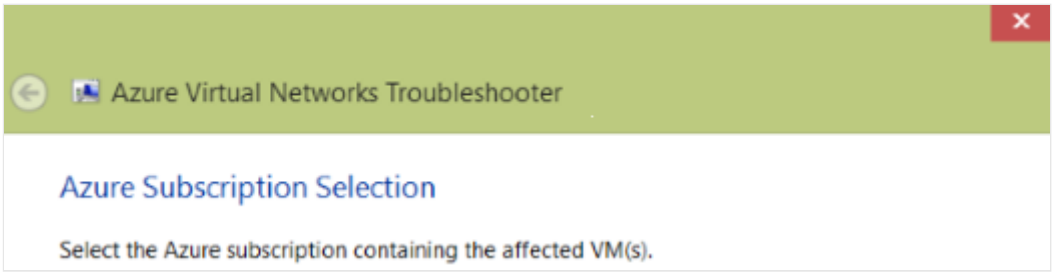
Execution details

The following steps and screenshots provide a walk-through of the diagnostic package:

1. Provide your credentials to authenticate to the affected Azure subscription.

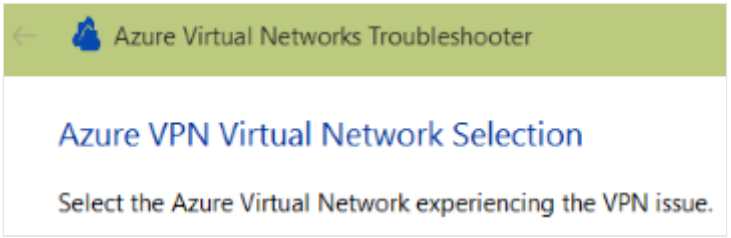


2. Confirm the Azure subscription name.

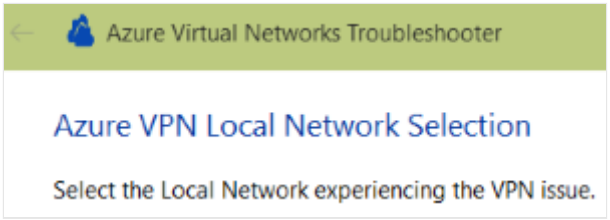


3. Select the scenario that best fits the issue that is being experienced.

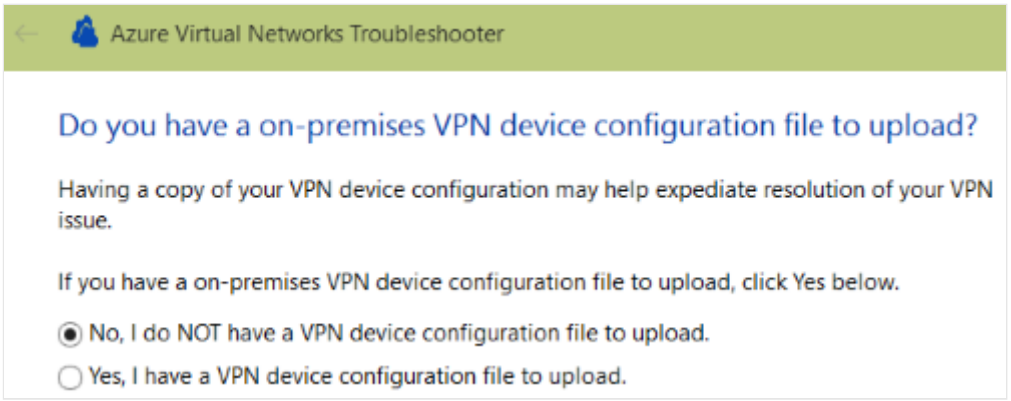
4. Select the affected Azure virtual network.

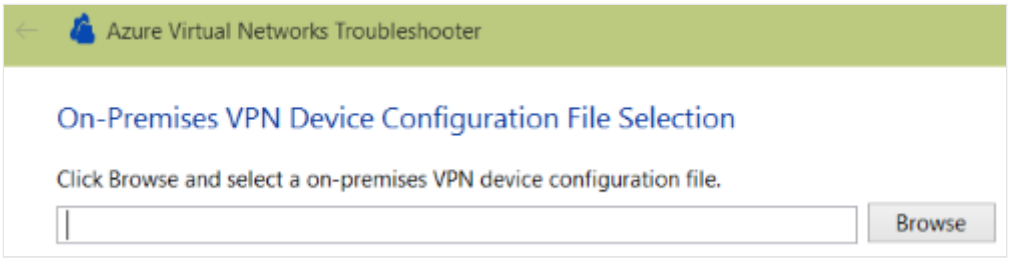


5. Select a local network.

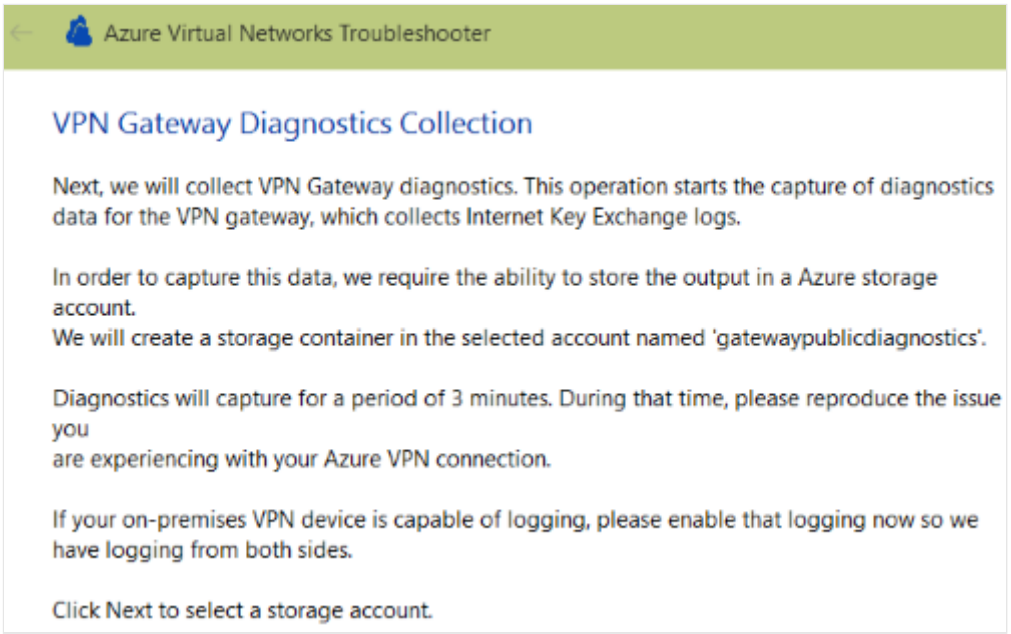


6. Respond to the prompt for an on-premises device configuration file.

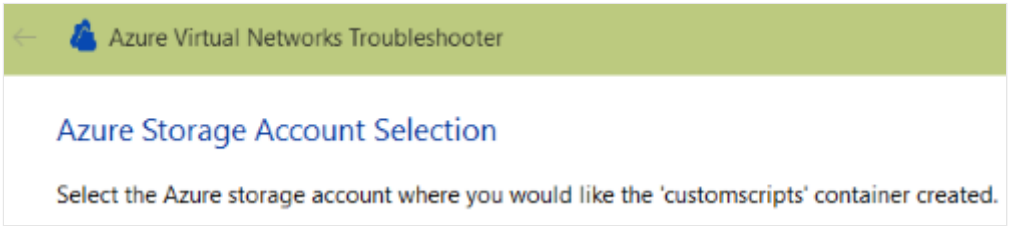




7. Notify the customer about Azure Gateway diagnostics collection.

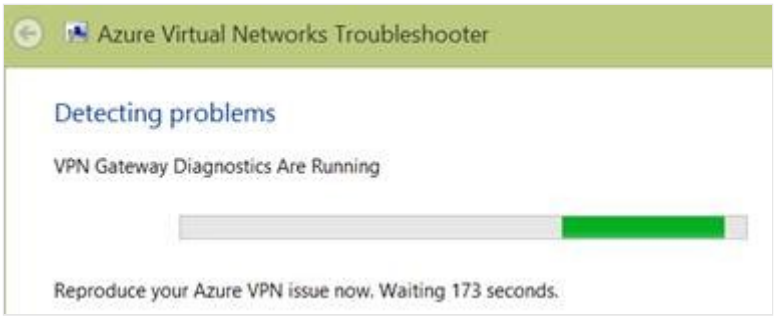


8. Respond to the prompt for storage account selection.



9. Display information that shows that the diagnostics are currently running, and ask the customer to reproduce the VPN issue.

   (Seconds count down on the screen.)



10. Respond to the prompt for an on-premises device log file.

Contact us for help

If you have questions or need help, create a support request, or ask Azure community support. You can also submit product feedback to Azure feedback community.

Feedback

Was this page helpful? Yes No

Configure reverse lookup zones for an SMTP banner check

Summarize this article for me

Summary

This article describes how to use a reverse zone in Azure DNS and create a Reverse DNS (PTR) record for SMTP Banner Check.

Symptom

If you host an SMTP server in Microsoft Azure, you may receive the following error message when send or receive a message from remote mail servers:

554: No PTR Record

Solution

For a virtual IP address in Azure, the reverse records are created in Microsoft owned domain zones, not custom domain zones.

To configure PTR records in Microsoft owned zones, use the -ReverseFqdn property on the PublicIpAddress resource. For more information, see Configure reverse DNS for services hosted in Azure.

When you configure the PTR records, make sure that the IP address and the reverse FQDN are owned by the subscription. If you try to set a reverse FQDN that does not belong to the subscription, you receive the following error message:

Output

```
Set-AzPublicIpAddress : ReverseFqdn mail.contoso.com that PublicIPAddress ip01 is
trying to use does not belong to subscription <Subscription ID>. One of the
following conditions need to be met to establish ownership:

1) ReverseFqdn matches fqdn of any public ip resource under the subscription;
2) ReverseFqdn resolves to the fqdn (through CName records chain) of any public ip
resource under the subscription;
3) It resolves to the ip address (through CName and A records chain) of a static
public ip resource under the subscription.
```

If you manually change your SMTP banner to match our default reverse FQDN, the remote mail server can still fail because it may expect the SMTP banner host to match the MX record for the domain.

Last updated on 03/02/2026

Diagnose a virtual machine routing problem

Summarize this article for me

Summary

In this article, you learn how to diagnose routing problems by viewing the effective routes for a network interface in a virtual machine (VM). Azure automatically creates default routes for each virtual network subnet. You can override these default routes by defining custom routes in a route table and associating the table with a subnet. The effective routes for a network interface are a combination of Azure's default routes, custom routes you define, and any routes propagated from your on-premises network through an Azure VPN gateway using the border gateway protocol (BGP). If you're new to virtual networks, network interfaces, or routing, see Virtual network overview, Network interface, and Routing overview.

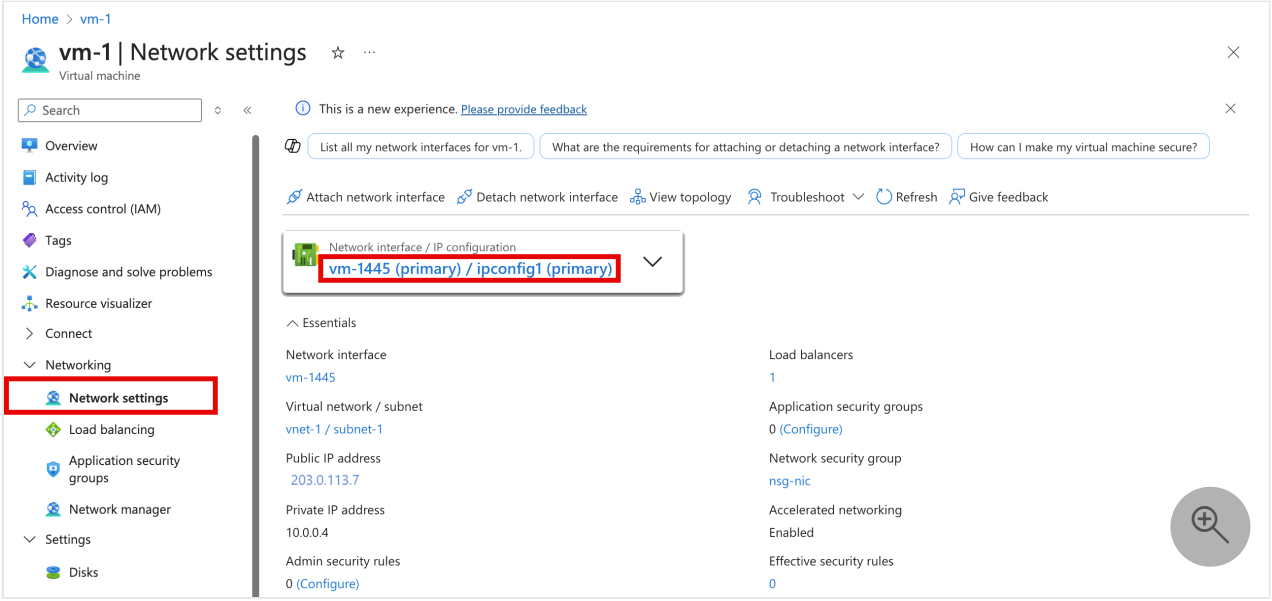
Scenario

You attempt to connect to a VM, but the connection fails. To determine why you can't connect to the VM, you can view the effective routes for a network interface using the Azure portal, PowerShell, or the Azure CLI.

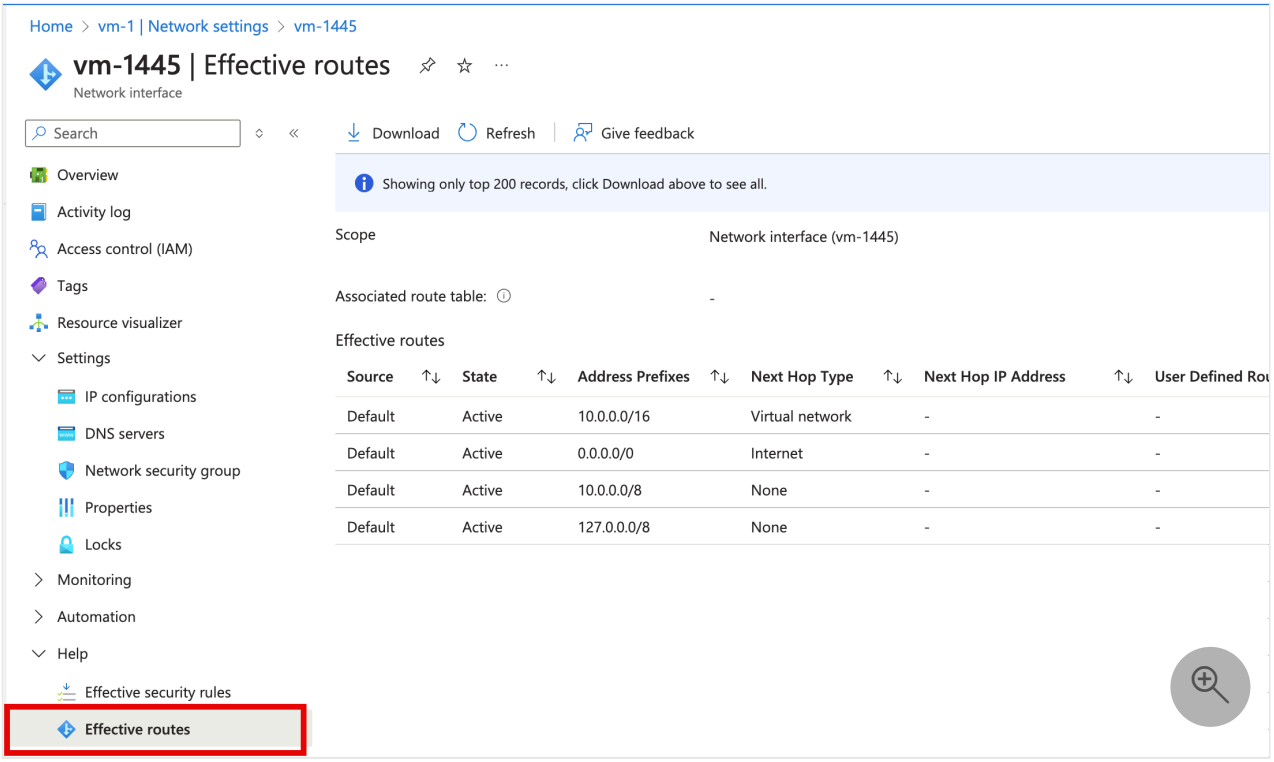
The steps that follow assume you have an existing VM to view the effective routes for. If you don't have an existing VM, first deploy a Linux or Windows VM to complete the tasks in this article with. The examples in this article are for a VM named *vm-1* with a network interface named *vm-1445*. The VM and network interface are in a resource group named *test-rg*, and are in the *East US* region. Change the values in the steps, as appropriate, for the VM you're diagnosing the problem for.

Diagnose using Azure portal

1. Sign-in the Azure portal with an Azure account that has the necessary permissions.
2. At the top of the Azure portal, enter the name of a VM that is in the running state, in the search box. When the name of the VM appears in the search results, select it.
3. Expand the **Networking** section and select **Network settings**.
4. To select the interface, select its name.



5. In the network interface, expand **Help**. Select **Effective routes**.



Select the desired network interface to view its effective routes. Each interface might belong to a different subnet, resulting in unique routes. The example in the image shows default routes created by Azure for each subnet. Your list includes the default routes and might also include extra routes. The routes could be from features like virtual network peering or connections to on-premises networks via an Azure VPN gateway. For details about the routes, see Virtual network traffic routing. If there are many routes, use the **Download** option to save them as a .csv file for easier review.

Though effective routes were viewed through the VM in the previous steps, you can also view effective routes through an:

- **Individual network interface**: Learn how to view a network interface.
- **Individual route table**: Learn how to view a route table.

Diagnose using PowerShell

> **Note**
>
> We recommend that you use the Azure Az PowerShell module to interact with Azure. To get started, see Install Azure PowerShell. To learn how to migrate to the Az PowerShell module, see Migrate Azure PowerShell from AzureRM to Az.

You can run the commands that follow in the Azure Cloud Shell, or by running PowerShell from your computer. The Azure Cloud Shell is a free interactive shell. It has common Azure tools preinstalled and configured to use with your account. If you run PowerShell from your computer, you need the Azure PowerShell module, version 1.0.0 or later. Run `Get-Module -ListAvailable Az` on your computer, to find the installed version. If you need to upgrade, see Install Azure PowerShell module. If you're running PowerShell locally, you also need to run `Connect-AzAccount` to log into Azure with an account that has the necessary permissions.

Get the effective routes for a network interface with Get-AzEffectiveRouteTable. The following example gets the effective routes for a network interface named *vm-1445* in a resource group named *test-rg*:

Azure PowerShell

```
$Params = @{
  NetworkInterfaceName = "vm-1445"
  ResourceGroupName    = "test-rg"
}
Get-AzEffectiveRouteTable @Params | Format-Table
```

To understand the information returned in the output, see Routing overview. Output is only returned if the VM is in the running state. If there are multiple network interfaces attached to the VM, you can review the effective routes for each network interface. Since each network interface can be in a different subnet, each network interface can have different effective routes. If you're still having a communication problem, see more diagnoses and considerations.

If you know the VM name but not the network interface name, use the following commands to return the ID of all network interfaces attached to the VM:

Azure PowerShell

```
$Params = @{
  Name               = "vm-1"
  ResourceGroupName = "test-rg"
}
$VM = Get-AzVM @Params
$VM.NetworkProfile
```

You receive output similar to the following example:

Output

```
NetworkInterfaces
-----------------
{/subscriptions/<ID>/resourceGroups/test-
rg/providers/Microsoft.Network/networkInterfaces/vm-1445
```

In the previous output, the network interface name is *vm-1445*.

Diagnose using Azure CLI

You can run the commands that follow in the Azure Cloud Shell, or by running the CLI from your computer. This article requires the Azure CLI version 2.0.32 or later. Run `az --version` to find the installed version. If you need to install or upgrade, see Install Azure CLI. If you're running the Azure CLI locally, you also need to run `az login` and log into Azure with an account that has the necessary permissions.

Get the effective routes for a network interface with az network nic show-effective-route-table. The following command gets the effective routes for a network interface named *vm-1445* that is in a resource group named *test-rg*:

Azure CLI

```
az network nic show-effective-route-table \
  --name vm-1445 \
  --resource-group test-rg
```

To understand the information returned in the output, see Routing overview. Output is only returned if the VM is in the running state. If there are multiple network interfaces attached to the VM, you can review the effective routes for each network interface. Since each network interface can be in a different subnet, each network interface can have different effective routes. If you're still having a communication problem, see more diagnoses and considerations.

If you know the VM name but not the network interface name, use the following commands to return the ID of all network interfaces attached to the VM:

Azure CLI

```
az vm show \
  --name vm-1 \
  --resource-group test-rg
```

Resolve a problem

Resolving routing problems typically consists of the following procedures:

- Addition of a custom route to override one of Azure's default routes. Learn how to add a custom route.
- Changed or removed a custom route that might cause routing to an undesired location. Learn how to change or delete a custom route.
- Ensure that the route table that contains any custom routes defined is associated to the subnet the network interface is in. Learn how to associate a route table to a subnet.
- Ensure that devices such as Azure VPN gateway or network virtual appliances deployed are operable. Use the VPN diagnostics capability of Network Watcher to determine any problems with an Azure VPN gateway.

If you're still having communication problems, see Considerations and more diagnoses.

Considerations

Consider the following points when troubleshooting communication problems:

- Routing uses the longest prefix match (LPM) to determine the best route from system routes, BGP, and custom routes. If multiple routes share the same LPM match, Azure selects one based on the priority order in Routing overview. Effective routes show only the LPM-matched routes, making it easier to identify and troubleshoot routes affecting VM communication.
- If custom routes direct traffic to a network virtual appliance (NVA) with *Virtual Appliance* as the next hop type, ensure the NVA's IP forwarding is enabled; otherwise, packets are dropped. Learn how to enable IP forwarding for a network interface and configure the NVA's operating system or application to forward traffic.
- If a route to 0.0.0.0/0 is created, all outbound internet traffic is routed to the next hop you specified, such as to an NVA or VPN gateway. Creating such a route is often referred to as forced tunneling. Remote connections using the RDP or SSH protocols from the internet

to your VM might not work with this route, depending on how the next hop handles the traffic. Forced-tunneling can be enabled:
  - When using site-to-site VPN, by creating a route with a next hop type of *VPN Gateway*. Learn more about configuring forced tunneling.
  - If a 0.0.0.0/0 (default route) is advertised over BGP through a virtual network gateway when using a site-to-site VPN, or ExpressRoute circuit. Learn more about using BGP with a site-to-site VPN or ExpressRoute.

- For virtual network peering traffic to work correctly, a system route with a next hop type of *VNet Peering* must exist for the peered virtual network's prefix range. If such a route doesn't exist, and the virtual network peering link is **Connected**:

  - Wait a few seconds, and retry. If it's a newly established peering link, it occasionally takes longer to propagate routes to all the network interfaces in a subnet. To learn more about virtual network peering, see Virtual network peering overview and manage virtual network peering.

  - Network security group rules might be impacting communication. For more information, see Diagnose a virtual machine network traffic filter problem.

- Though Azure assigns default routes to each Azure network interface, if you have multiple network interfaces attached to the VM, only the primary network interface is assigned a default route (0.0.0.0/0), or gateway, within the VM's operating system. Learn how to create a default route for secondary network interfaces attached to a Windows or Linux VM. Learn more about primary and secondary network interfaces.

More diagnoses

- To run a quick test to determine the next hop type for traffic destined to a location, use the Next hop capability of Azure Network Watcher. Next hop tells you what the next hop type is for traffic destined to a specified location.

- If there are no routes causing a VM network communication to fail, the problem might be due to firewall software running within the VM's operating system

- If you're force tunneling traffic to an on-premises device through a VPN gateway, or NVA, you might not be able to connect to a VM from the internet, depending on how routing is configured for the devices. Confirm that the routing configured for the device routes traffic to either a public or private IP address for the VM.

- Use the connection troubleshoot capability of Network Watcher to determine routing, filtering, and in-OS causes of outbound communication problems.

Next steps

- Learn about all tasks, properties, and settings for a route table and routes.
- Learn about all next hop types, system routes, and how Azure selects a route.

Last updated on 03/02/2026

Diagnose a virtual machine network traffic filter problem

Summarize this article for me

Summary

In this article, you learn how to diagnose network traffic filter issues for a virtual machine (VM). The process involves viewing the effective security rules applied by the network security group (NSG).

NSGs enable you to control the types of traffic that flow in and out of a VM. You can associate an NSG to a subnet in an Azure virtual network, a network interface attached to a VM, or both. The effective security rules applied to a network interface are an aggregation of the rules that exist in the NSG associated to a network interface, and the subnet the network interface is in. Rules in different NSGs can sometimes conflict with each other and affect a VM's network connectivity. You can view all the effective security rules from NSGs that are applied on your VM's network interfaces. If you're not familiar with virtual network, network interface, or NSG concepts, see Virtual network overview, Network interface, and Network security groups overview.

Scenario

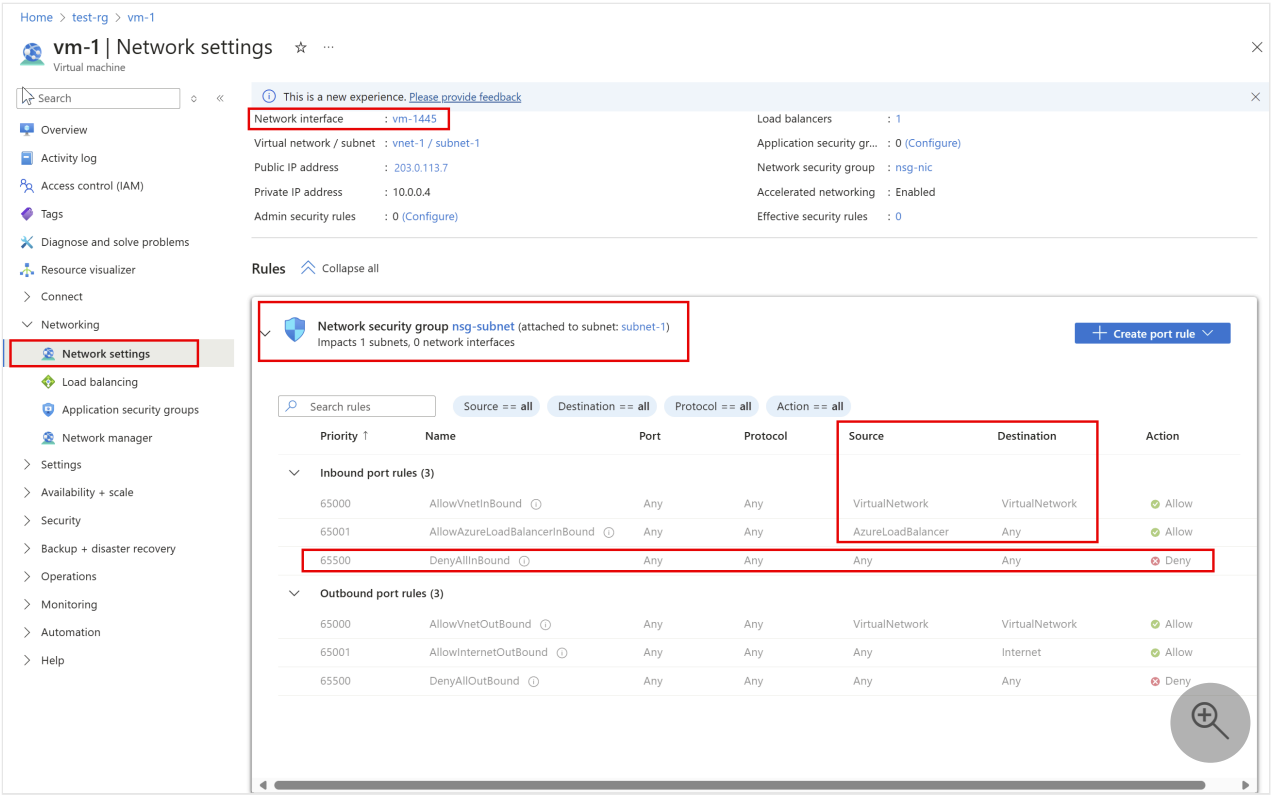
You attempt to connect to a VM over port 80 from the internet, but the connection fails. To determine why you can't access port 80 from the Internet, you can view the effective security rules for a network interface using the Azure portal, PowerShell, or the Azure CLI.

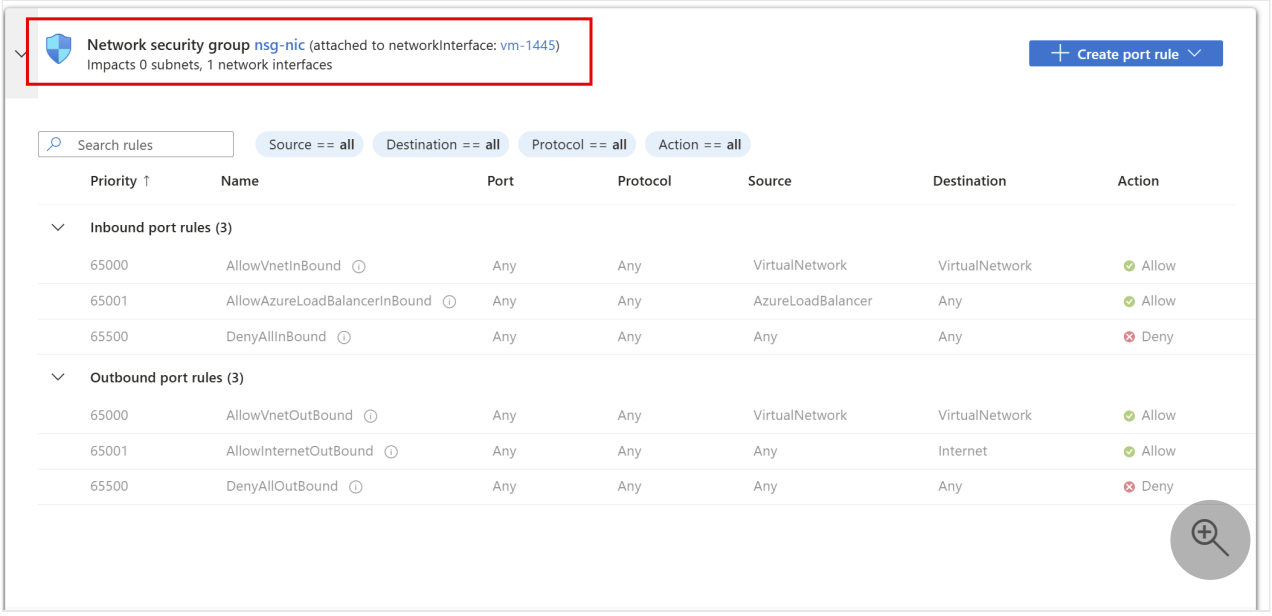
The steps that follow assume you have an existing VM to view the effective security rules for. If you don't have an existing VM, first deploy a Linux or Windows VM to complete the tasks in this article with. The examples in this article are for a VM named *myVM* with a network interface named *myVMVMNic*. The VM and network interface are in a resource group named *myResourceGroup*, and are in the *East US* region. Change the values in the steps, as appropriate, for the VM you're diagnosing the problem for.

Diagnose using Azure portal

1. Log into the Azure portal with an Azure account that has the necessary permissions.
2. At the top of the Azure portal, enter the name of the VM in the search box. When the name of the VM appears in the search results, select it.

3. Expand **Networking** in the left pane. Select **Network settings**. The following figures show the network security group settings for the VM's network interface.





The rules you see listed in the previous figures are for a network interface named **vm-1445**. You see that there are **Inbound port rules** for the network interface from two different network security groups:

- **nsg-subnet**: Associated to the subnet that the network interface is in.
- **nsg-nic**: Associated to the network interface in the VM named **vm-1445**.

The rule named **DenyAllInBound** is what's preventing inbound communication to the VM over port 80, from the internet, as described in the scenario. The rule lists *0.0.0.0/0* for **SOURCE**, which includes the internet. No other rule with a higher priority (lower number)

allows port 80 inbound. To allow port 80 inbound to the VM from the internet, see Resolve a problem. To learn more about security rules and how Azure applies them, see Network security groups.

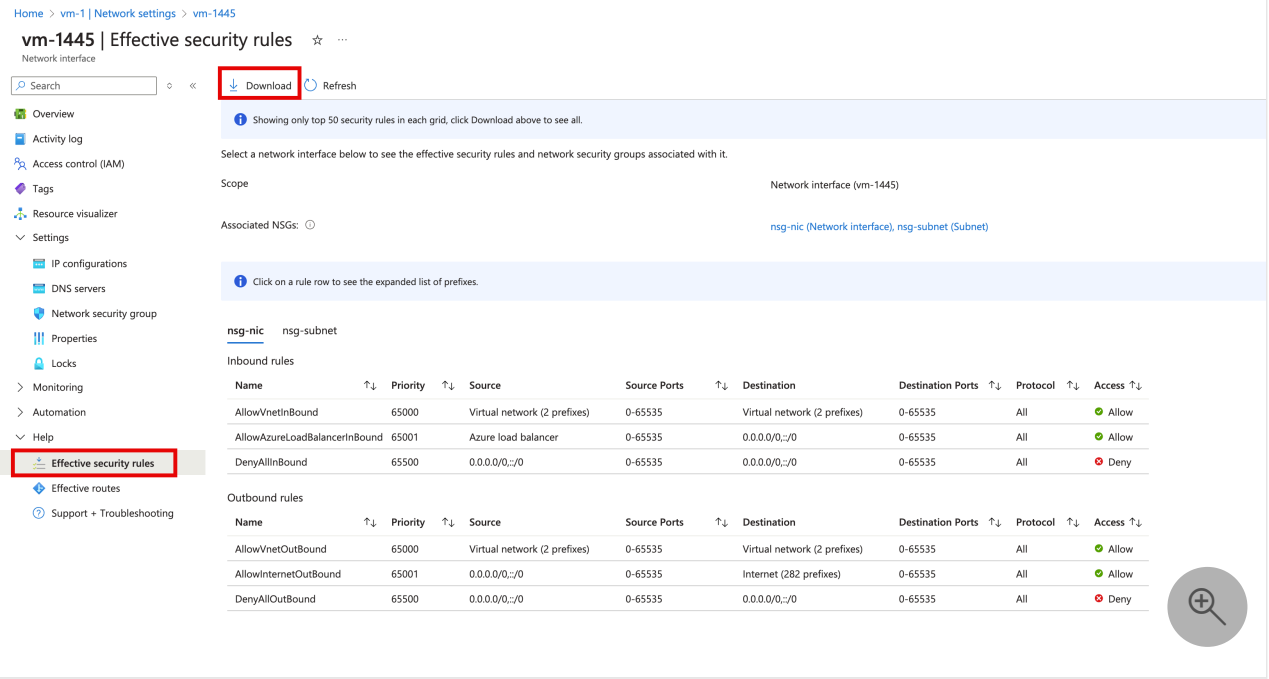
At the bottom of the picture, you also see **Outbound port rules**. The outbound port rules for the network interface are listed.

Though the picture only shows four inbound rules for each NSG, your NSGs might have many more than four rules. In the picture, you see **VirtualNetwork** under **Source** and **Destination** and **AzureLoadBalancer** under **SOURCE**. **VirtualNetwork** and **AzureLoadBalancer** are service tags. Service tags represent a group of IP address prefixes to help minimize complexity for security rule creation.

4. To view the effective security rules, select the interface in the network settings of the virtual machine. Ensure the VM is in a running state before proceeding.

5. In the settings for the network interface, expand **Help**, then select **Effective security rules**.

   The following example shows the example network interface **vm-1445** with the **Effective security rules** selected.

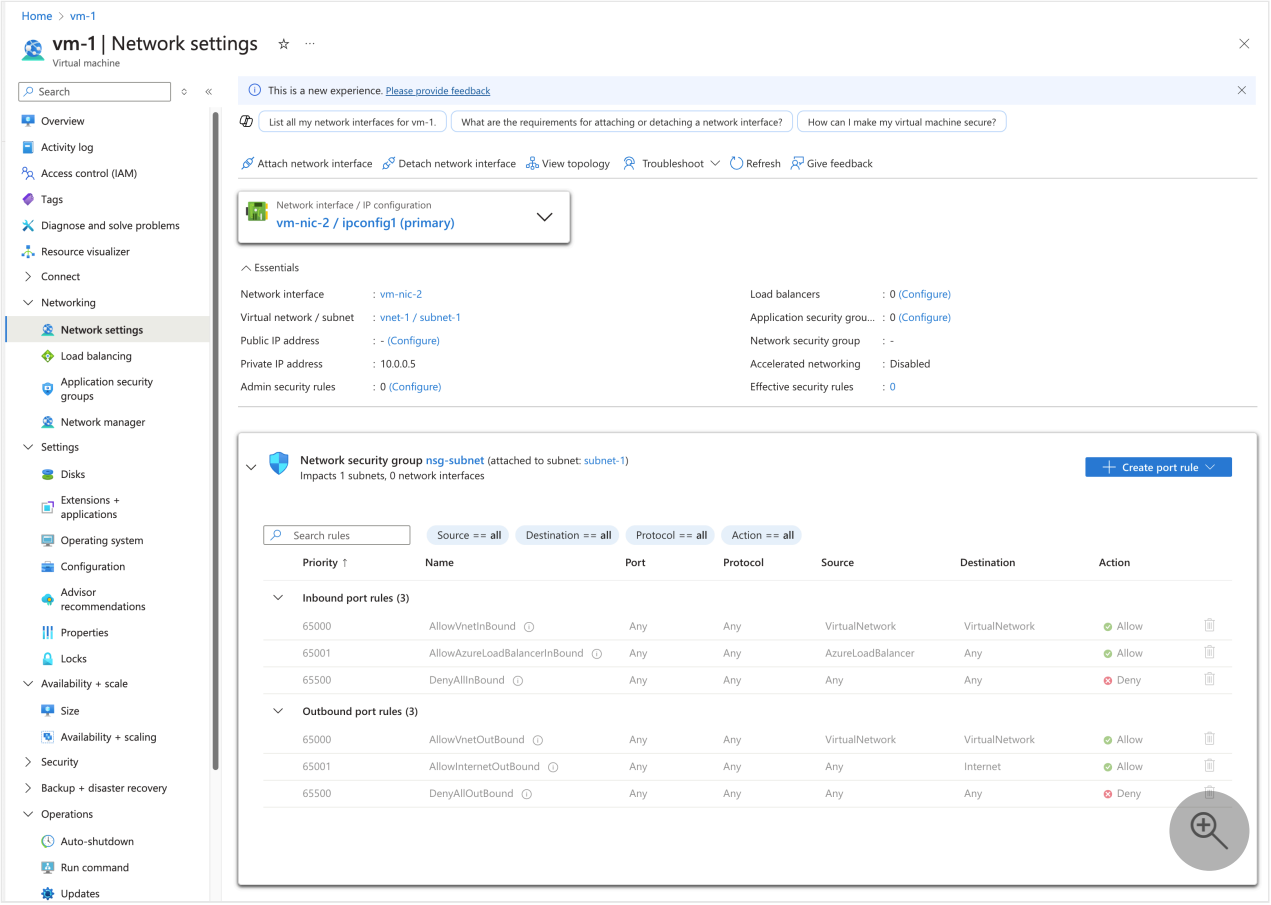


   The rules listed are the same as you saw in step 3, though there are different tabs for the NSG associated to the network interface and the subnet. As you can see in the picture, only the first 50 rules are shown. To download a .csv file that contains all of the rules, select **Download**.

6. The previous steps showed the security rules for a network interface named **vm-1445**. What if a VM has two network interfaces? The VM in this example has two network

interfaces attached to it. The effective security rules can be different for each network interface.

To see the rules for the **vm-nic-2** network interface, select it. As shown in the example that follows, the network interface has the same rules associated to its subnet as the **vm-1445** network interface, because both network interfaces are in the same subnet. When you associate an NSG to a subnet, its rules are applied to all network interfaces in the subnet.



Unlike the **vm-1445** network interface, the **vm-nic-2** network interface doesn't have a network security group associated to it. Each network interface and subnet can have zero, or one, NSG associated to it. The NSG associated to each network interface or subnet can be the same, or different. You can associate the same network security group to as many network interfaces and subnets as you choose.

Though effective security rules were viewed through the VM, you can also view effective security rules through an individual:

- **Network interface**: Learn how to view a network interface.
- **NSG**: Learn how to view an NSG.

Diagnose using PowerShell

> **Note**
>
> We recommend that you use the Azure Az PowerShell module to interact with Azure. To get started, see Install Azure PowerShell. To learn how to migrate to the Az PowerShell module, see Migrate Azure PowerShell from AzureRM to Az.

You can run the commands that follow in the Azure Cloud Shell, or by running PowerShell from your computer. The Azure Cloud Shell is a free interactive shell. It has common Azure tools preinstalled and configured to use with your account. If you run PowerShell from your computer, you need the Azure PowerShell module, version 1.0.0 or later. Run `Get-Module -ListAvailable Az` on your computer, to find the installed version. If you need to upgrade, see Install Azure PowerShell module. If you're running PowerShell locally, you also need to run `Connect-AzAccount` to log into Azure with an account that has the necessary permissionsx.

Get the effective security rules for a network interface with Get-AzEffectiveNetworkSecurityGroup. The following example gets the effective security rules for a network interface named *vm-nic* that is in a resource group named *test-rg*:

Azure PowerShell

```
$Params = @{
  NetworkInterfaceName = "vm-nic"
  ResourceGroupName    = "test-rg"
}
Get-AzEffectiveNetworkSecurityGroup @Params
```

Output is returned in json format. To understand the output, see interpret command output. Output is only returned if an NSG is associated with the network interface, the subnet the network interface is in, or both. The VM must be in the running state. A VM might have multiple network interfaces with different NSGs applied. When troubleshooting, run the command for each network interface.

If you're still having a connectivity problem, see more diagnoses and considerations.

If you only know the VM name, use the following commands to list all network interface IDs attached to the VM.

Azure PowerShell

```
$Params = @{
  Name               = "vm-1"
  ResourceGroupName = "test-rg"
}
```

```
$VM = Get-AzVM @Params
$VM.NetworkProfile
```

You receive output similar to the following example:

Output

```
NetworkInterfaces
-----------------
{/subscriptions/<ID>/resourceGroups/myResourceGroup/providers/Microsoft.Network/net
workInterfaces/vm-nic
```

In the previous output, the network interface name is *vm-nic*.

Diagnose using Azure CLI

If using Azure CLI commands to complete tasks in this article, either run the commands in the Azure Cloud Shell, or by running the Azure CLI from your computer. This article requires the Azure CLI version 2.0.32 or later. Run `az --version` to find the installed version. If you need to install or upgrade, see Install Azure CLI. If you're running the Azure CLI locally, you also need to run `az login` and log into Azure with an account that has the necessary permissions.

Get the effective security rules for a network interface with az network nic list-effective-nsg. The following example gets the effective security rules for a network interface named *vm-nic* that is in a resource group named *test-rg*:

Azure CLI

```
az network nic list-effective-nsg \
  --name vm-nic \
  --resource-group test-rg
```

Output is returned in json format. To understand the output, see interpret command output. Output is only returned if an NSG is associated with the network interface, the subnet the network interface is in, or both. The VM must be in the running state. A VM might have multiple network interfaces with different NSGs applied. When troubleshooting, run the command for each network interface.

If you're still having a connectivity problem, see more diagnoses and considerations.

If you only know the VM name, use the following commands to list all network interface IDs attached to the VM.

Azure CLI

```
az vm show \
  --name vm-1 \
  --resource-group test-rg
```

Within the returned output, you see information similar to the following example:

Output

```
"networkProfile": {
    "additionalProperties": {},
    "networkInterfaces": [
      {
        "additionalProperties": {},
        "id": "/subscriptions/<ID>/resourceGroups/test-
rg/providers/Microsoft.Network/networkInterfaces/vm-nic",
        "primary": true,
        "resourceGroup": "test-rg"
      },
```

In the previous output, the network interface name is **vm-nic**

Interpret command output

Regardless of whether you used the PowerShell, or the Azure CLI to diagnose the problem, you receive output that contains the following information:

- **NetworkSecurityGroup**: The ID of the network security group.
- **Association**: Indicates whether the network security group is linked to a *NetworkInterface* or *Subnet*. If the NSG's association status changes right before running the command, wait a few seconds for the output to reflect the update.
- **EffectiveSecurityRules**: An explanation of each property is detailed in Create a security rule. Rule names prefaced with *defaultSecurityRules/* are default security rules that exist in every NSG. Rule names prefaced with *securityRules/* are user created rules. Rules that specify a service tag, such as **Internet**, **VirtualNetwork**, and **AzureLoadBalancer** for the **destinationAddressPrefix** or **sourceAddressPrefix** properties, also have values for the **expandedDestinationAddressPrefix** property. The **expandedDestinationAddressPrefix** property lists all address prefixes represented by the service tag.

Duplicate rules appear in the output when an NSG is linked to both the network interface and the subnet. Default rules and any custom rules shared between the NSGs cause these duplicates.

The rule named **defaultSecurityRules/DenyAllInBound** is what's preventing inbound communication to the VM over port 80, from the internet, as described in the scenario. No

other rule with a higher priority (lower number) allows port 80 inbound from the internet.

Resolve a problem

You can diagnose the problem described in the scenario using the Azure portal, PowerShell, or the Azure CLI. The solution is to create a network security rule with the following properties:

Expand table

Property	Value
Source	Any
Source port ranges	Any
Destination	The IP address of the VM, a range of IP addresses, or all addresses in the subnet.
Destination port ranges	80
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-HTTP-All

After rule creation, port 80 is allowed inbound from the internet because its priority is higher than the default *DenyAllInBound* rule. If NSGs are associated with both the network interface and the subnet, create the same rule in both NSGs. Learn how to create a security rule.

When Azure processes inbound traffic, it processes rules in the NSG associated to the subnet (if there's an associated NSG), and then it processes the rules in the NSG associated to the network interface. If there's an NSG associated to the network interface and the subnet, the port must be open in both NSGs, for the traffic to reach the VM. To ease administration and communication problems, we recommend that you associate an NSG to a subnet, rather than individual network interfaces. If VMs within a subnet need different security rules, you can make the network interfaces members of an application security group (ASG), and specify an ASG as the source and destination of a security rule. Learn more about application security groups.

If you're still having communication problems, see Considerations and more diagnosis.

Considerations

Consider the following points when troubleshooting connectivity problems:

- Default security rules block inbound access from the internet, and only permit inbound traffic from the virtual network. To allow inbound traffic from the Internet, add security rules with a higher priority than default rules. Learn more about default security rules, or how to add a security rule.

- For peered virtual networks, by default, the **VIRTUAL_NETWORK** service tag automatically expands to include prefixes for peered virtual networks. To troubleshoot any issues related to virtual network peering, you can view the prefixes in the **ExpandedAddressPrefix** list. Learn more about virtual network peering and service tags.

- Azure only shows effective security rules for a network interface when an NSG is associated with the VM's network interface or subnet. Additionally, the VM must be in the running state.

- If the network interface or subnet lacks an associated NSG, all ports remain open for inbound and outbound access when a VM has a public IP address. This configuration allows unrestricted access to and from anywhere. To secure the VM, apply an NSG to the subnet hosting the network interface if it has a public IP address.

More diagnoses

- To run a quick test to determine if traffic is allowed to or from a VM, use the IP flow verify capability of Azure Network Watcher. IP flow verify tells you if traffic is allowed or denied. If denied, IP flow verify tells you which security rule is denying the traffic.
- If there are no security rules causing a VM's network connectivity to fail, the problem might be due to:
  - Firewall software running within the VM's operating system
  - Routes configured for virtual appliances or on-premises traffic. Internet traffic can be redirected to your on-premises network via forced-tunneling. If you force tunnel internet traffic to a virtual appliance, or on-premises, you might not be able to connect to the VM from the internet. To learn how to diagnose route problems that might impede the flow of traffic out of the VM, see Diagnose a virtual machine network traffic routing problem.

Next steps

- Learn about all tasks, properties, and settings for a network security group and security rules.
- Learn about default security rules, service tags, and how Azure processes security rules for inbound and outbound traffic for a VM.

Last updated on 03/02/2026

Network virtual appliance issues in Azure

Summarize this article for me

> **Note**
>
> We recommend that you use the Azure Az PowerShell module to interact with Azure. To get started, see Install Azure PowerShell. To learn how to migrate to the Az PowerShell module, see Migrate Azure PowerShell from AzureRM to Az.

Summary

You might experience VM or VPN connectivity issues and errors when using a partner Network Virtual Appliance (NVA) in Microsoft Azure. This article provides basic steps to help you validate basic Azure Platform requirements for NVA configurations.

Technical support for partner NVAs and their integration with the Azure platform is provided by the NVA vendor.

> **Note**
>
> If you have a connectivity or routing problem that involves an NVA, you should contact the vendor of the NVA directly.

If your Azure issue is not addressed in this article, visit the Azure forums on Microsoft Q & A and Stack Overflow. You can post your issue in these forums, or post to @AzureSupport on Twitter. You also can submit an Azure support request. To submit a support request, on the Azure support page, select **Get support**.

Checklist for troubleshooting with NVA vendor

- Software updates for NVA VM software
- Service Account setup and functionality
- User-defined routes (UDRs) on virtual network subnets that direct traffic to NVA
- UDRs on virtual network subnets that direct traffic from NVA
- Routing tables and rules within the NVA (for example, from NIC1 to NIC2)

- Tracing on NVA network interfaces to verify receiving and sending network traffic
- Use of a Standard version Public IP. There must be an NSG created and an explicit rule to allow the traffic to be routed to the NVA.

Basic troubleshooting steps

- Check the basic configuration
- Check NVA performance
- Advanced network troubleshooting

Check the minimum configuration requirements for NVAs on Azure

Each NVA has basic configuration requirements to function correctly on Azure. The following section provides the steps to verify these basic configurations. For more information, contact the vendor of the NVA.

Check whether IP forwarding is enabled on NVA

Portal

1. Locate the NVA resource in the Azure portal, select Networking, and then select the Network interface.
2. On the Network interface page, select IP configuration.
3. Ensure that the IP forwarding checkbox is selected.

Check for NSG when using Standard SKU public IP

Use of a standard version of public IPs. There must be an NSG created and an explicit rule to allow the traffic to the NVA.

Check whether the traffic can be routed to the NVA

1. On Azure portal, open **Network Watcher**, select **Next Hop**.
2. Specify a VM that is configured to redirect the traffic to the NVA, and a destination IP address at which to view the next hop.

3. If the NVA isn't listed as the **next hop**, check and update the Azure route tables.

Check whether the traffic can reach the NVA

1. In Azure portal, open **Network Watcher**, and then select **IP Flow Verify**.
2. Specify the VM and the IP address of the NVA. Check for traffic blockage by any Network security groups (NSG).
3. If there's an NSG rule that blocks the traffic, locate the NSG in **effective security** rules and then update it to allow traffic to pass. Then run **IP Flow Verify** again and use **Connection troubleshoot** to test TCP communications from VM to your internal or external IP address.

Check whether NVA and VMs are listening for expected traffic

Connect to the NVA by using RDP or SSH, and then run following command:

For Windows:

Console

```
netstat -an
```

For Linux:

Console

```
netstat -an | grep -i listen
```

If the TCP port used by the NVA software isn't listed in the results, configure the application on the NVA and VM to listen on those ports. For further assistance, contact the NVA vendor.

Check NVA performance

Validate VM CPU

If CPU usage gets close to 100 percent, you might experience issues that affect network packet drops. Your VM reports average CPU for a specific time span in the Azure portal. During a CPU spike, investigate which process on the guest VM is causing the high CPU, and mitigate it, if possible. You might also have to resize the VM to a larger SKU size or, for virtual machine scale set, increase the instance count or set to autoscale on CPU usage. For either of these issues, contact the NVA vendor for assistance, as needed.

Validate VM network statistics

If the VM network use spikes or shows periods of high usage, you might also have to increase the SKU size of the VM to obtain higher throughput capabilities. You can also redeploy the VM by having Accelerated Networking enabled. To verify whether the NVA supports Accelerated Networking feature, contact the NVA vendor for assistance, as needed.

Advanced network administrator troubleshooting

Capture network trace

Capture a simultaneous network trace on the source VM, the NVA, and the destination VM while you run **PsPing** or **Nmap**, and then stop the trace.

1. To capture a simultaneous network trace, run the following command:

   For Windows

   Console
   ```
   netsh trace start capture=yes tracefile=c:\server_IP.etl
   scenario=netconnection
   ```

   For Linux

   Console
   ```
   sudo tcpdump -s0 -i eth0 -X -w vmtrace.cap
   ```

2. Use **PsPing** or **Nmap** from the source VM to the destination VM (for example: `PsPing 10.0.0.4:80` or `Nmap -p 80 10.0.0.4`).

3. Open the network trace from the destination VM by using Network Monitor or tcpdump. Apply a display filter for the IP of the Source VM you ran **PsPing** or **Nmap** from, such as `IPv4.address==10.0.0.4 (Windows netmon)` or `tcpdump -nn -r vmtrace.cap src or dst host 10.0.0.4` (Linux).

Analyze traces

If you don't see the packets incoming to the backend VM trace, there's likely an NSG or UDR interfering or the NVA routing tables are incorrect.

If you do see the packets coming in but no response, then you might need to address a VM application or a firewall issue. For either of these issues, contact the NVA vendor for assistance as needed.

Last updated on 03/02/2026

Resource logging for a network security group

Summarize this article for me

Summary

A network security group (NSG) includes rules that allow or deny traffic to a virtual network subnet, network interface, or both.

When you enable logging for an NSG, you can gather the following types of resource log information:

- **Event**: Entries are logged for the NSG rules applied to virtual machines, based on a MAC address.
- **Rule counter**: Tracks how many times each NSG rule is applied to allow or deny traffic. The status for these rules is collected every 300 seconds.

Resource logs are available only for NSGs deployed through the Azure Resource Manager (ARM) deployment model. You can't enable resource logging for NSGs deployed through the classic deployment model. For more information, see Understand deployment models.

Resource logging is enabled separately for *each* NSG from which you want to collect diagnostic data. If you're interested in *activity* or *operational* logs instead, see Overview of Azure platform logs. If you're interested in IP traffic flowing through the virtual network, see Virtual network flow logs.

Enable logging

You can use the Azure portal, Azure PowerShell, or the Azure CLI to enable resource logging.

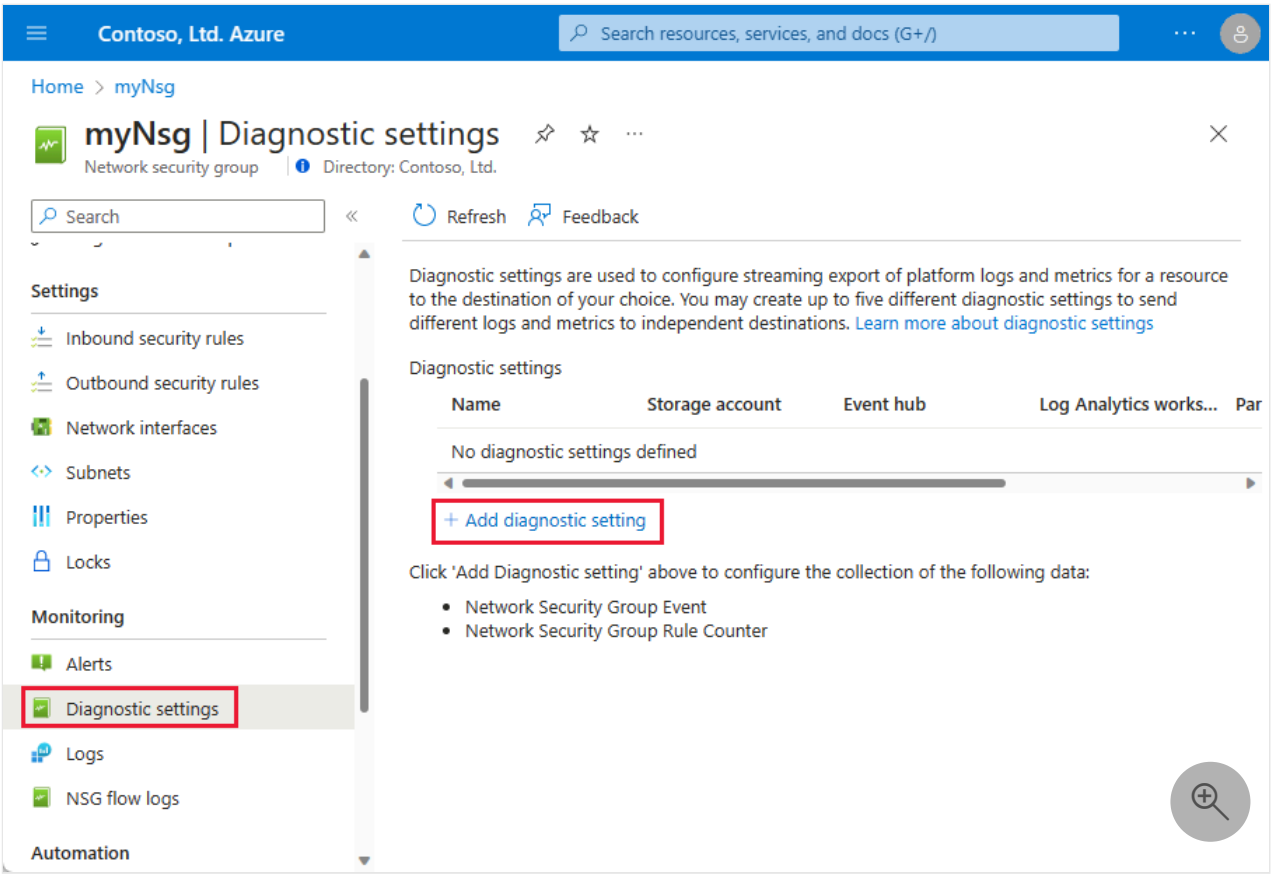
Azure portal

1. Sign in to the Azure portal.

2. In the search box at the top of the Azure portal, enter **network security groups**. Select **Network security groups** in the search results.

3. Select the NSG for which you want to enable logging.

4. Under **Monitoring**, select **Diagnostic settings**, then select **Add diagnostic setting**.



5. In **Diagnostic setting**, enter a name, such as **myNsgDiagnostic**.

6. For **Logs**, select **allLogs** or select individual categories of logs. For more information about each category, see Log categories.

7. Under **Destination details**, select one or more destinations:

   - Send to a Log Analytics workspace.
   - Archive to a storage account.
   - Stream to an event hub.
   - Send to a partner solution.

   For more information, see Log destinations.

8. Select **Save**.

9. View and analyze logs. For more information, see View and analyze logs.

Azure PowerShell

> ⓘ Note

We recommend that you use the Azure Az PowerShell module to interact with Azure. To get started, see Install Azure PowerShell. To learn how to migrate to the Az PowerShell module, see Migrate Azure PowerShell from AzureRM to Az.

You can run the commands that are available in this section in Azure Cloud Shell. You can also run PowerShell from your computer. Cloud Shell is a free interactive shell. It has common Azure tools preinstalled and configured to use with your account.

If you run PowerShell from your computer, you need the Azure PowerShell module, version 1.0.0 or later. Run `Get-Module -ListAvailable Az` to find the installed version. If you need to upgrade, see Install Azure PowerShell module. If you run PowerShell locally, you also need to run the Connect-AzAccount cmdlet to sign in to Azure with an account that has the necessary permissions.

To enable resource logging, you need the ID of an existing NSG. If you don't have an existing NSG, create one by using the New-AzNetworkSecurityGroup cmdlet.

Get the NSG for which you want to enable resource logging by using the Get-AzNetworkSecurityGroup cmdlet. Store the NSG in a variable for later use. For example, to retrieve an NSG named `myNsg` that exists in a resource group named `myResourceGroup`, enter the following command:

Azure PowerShell

```
$Nsg=Get-AzNetworkSecurityGroup `
  -Name myNsg `
  -ResourceGroupName myResourceGroup
```

You can write resource logs to different destination types. For more information, see Log destinations. In this article, logs are sent to a *Log Analytics workspace* destination. If you don't have an existing workspace, create one by using the New-AzOperationalInsightsWorkspace cmdlet.

Retrieve an existing Log Analytics workspace with the Get-AzOperationalInsightsWorkspace cmdlet. For example, to get and store an existing workspace named `myWorkspace` in a resource group named `myWorkspaces`, enter the following command:

Azure PowerShell

```
$Oms=Get-AzOperationalInsightsWorkspace `
  -ResourceGroupName myWorkspaces `
  -Name myWorkspace
```

There are two categories of logging that you can enable. For more information, see Log categories. Enable resource logging for the NSG with the New-AzDiagnosticSetting cmdlet. The following example logs both event and rule counter category data to the workspace for an NSG. It uses the IDs for the NSG and workspace that you obtained with the previous command:

Azure PowerShell

```
New-AzDiagnosticSetting `
   -Name myDiagnosticSetting `
   -ResourceId $Nsg.Id `
   -WorkspaceId $Oms.ResourceId
```

If you want to log to a different destination than a Log Analytics workspace, use an appropriate parameter in the command. For more information, see Azure resource logs.

For more information about settings, see New-AzDiagnosticSetting.

View and analyze logs. For more information, see View and analyze logs.

Azure CLI

You can run the commands in this section in Cloud Shell or by running the Azure CLI from your computer. Cloud Shell is a free interactive shell. It has common Azure tools preinstalled and configured to use with your account.

If you run the CLI from your computer, you need version 2.0.28 or later. Run `az --version` on your computer to find the installed version. If you need to upgrade, see Install the Azure CLI. If you run the CLI locally, you also need to run `az login` to sign in to Azure with an account that has the necessary permissions.

To enable resource logging, you need the ID of an existing NSG. If you don't have an existing NSG, create one by using az network nsg create.

Get and store the NSG for which you want to enable resource logging with az network nsg show. For example, to retrieve an NSG named `myNsg` that exists in a resource group named `myResourceGroup`, enter the following command:

Azure CLI

```
nsgId=$(az network nsg show \
  --name myNsg \
  --resource-group myResourceGroup \
  --query id \
  --output tsv)
```

You can write resource logs to different destination types. For more information, see Log destinations. In this article, logs are sent to a Log Analytics workspace destination as an example. For more information, see Log categories.

Enable resource logging for the NSG with az monitor diagnostic-settings create. The following example logs both event and rule counter category data to an existing workspace named `myWorkspace`, which exists in a resource group named `myWorkspaces`. It uses the ID of the NSG you obtained with the previous command:

Azure CLI

```
az monitor diagnostic-settings create \
  --name myNsgDiagnostics \
  --resource $nsgId \
  --logs '[ { "category": "NetworkSecurityGroupEvent", "enabled": true,
"retentionPolicy": { "days": 30, "enabled": true } }, { "category":
"NetworkSecurityGroupRuleCounter", "enabled": true, "retentionPolicy": { "days":
30, "enabled": true } } ]' \
  --workspace myWorkspace \
  --resource-group myWorkspaces
```

If you don't have an existing workspace, create one by using the Azure portal or Azure PowerShell. There are two categories of logging for which you can enable logs.

If you only want to log data for one category or the other, remove the category you don't want to log data for in the previous command. If you want to log to a different destination than a Log Analytics workspace, use an appropriate parameter. For more information, see Azure resource logs.

View and analyze logs. For more information, see View and analyze logs.

Log destinations

You can send diagnostics data to the following options:

- Log Analytics workspace
- Azure Event Hubs
- Azure Storage
- Azure Monitor partner integrations

Log categories

JSON-formatted data is written for the following log categories:

- Event
- Rule counter

Event

The event log contains information of NSG rules applied to virtual machines, based on a MAC address. The following data is logged for each event. In the following example, the data is logged for a virtual machine with the IP address 192.168.1.4 and a MAC address of 00-0D-3A-92-6A-7C.

JSON

```
{
    "time": "[DATE-TIME]",
    "systemId": "[ID]",
    "category": "NetworkSecurityGroupEvent",
    "resourceId": "/SUBSCRIPTIONS/[SUBSCRIPTION-ID]/RESOURCEGROUPS/[RESOURCE-GROUP-
NAME]/PROVIDERS/MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/[NSG-NAME]",
    "operationName": "NetworkSecurityGroupEvents",
    "properties": {
        "vnetResourceGuid":"[ID]",
        "subnetPrefix":"192.168.1.0/24",
        "macAddress":"00-0D-3A-92-6A-7C",
        "primaryIPv4Address":"192.168.1.4",
        "ruleName":"[SECURITY-RULE-NAME]",
        "direction":"[DIRECTION-SPECIFIED-IN-RULE]",
        "priority":"[PRIORITY-SPECIFIED-IN-RULE]",
        "type":"[ALLOW-OR-DENY-AS-SPECIFIED-IN-RULE]",
        "conditions":{
            "protocols":"[PROTOCOLS-SPECIFIED-IN-RULE]",
            "destinationPortRange":"[PORT-RANGE-SPECIFIED-IN-RULE]",
            "sourcePortRange":"[PORT-RANGE-SPECIFIED-IN-RULE]",
            "sourceIP":"[SOURCE-IP-OR-RANGE-SPECIFIED-IN-RULE]",
            "destinationIP":"[DESTINATION-IP-OR-RANGE-SPECIFIED-IN-RULE]"
            }
        }
}
```

Rule counter

The rule counter log contains information of each NSG rule applied to resources. The following example data is logged each time a rule is applied. In the following example, the data is logged for a virtual machine with the IP address 192.168.1.4 and a MAC address of 00-0D-3A-92-6A-7C.

JSON

```
{
    "time": "[DATE-TIME]",
    "systemId": "[ID]",
    "category": "NetworkSecurityGroupRuleCounter",
    "resourceId": "/SUBSCRIPTIONS/[SUBSCRIPTION ID]/RESOURCEGROUPS/[RESOURCE-GROUP-
NAME]/PROVIDERS/MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/[NSG-NAME]",
    "operationName": "NetworkSecurityGroupCounters",
    "properties": {
        "vnetResourceGuid":"[ID]",
        "subnetPrefix":"192.168.1.0/24",
        "macAddress":"00-0D-3A-92-6A-7C",
        "primaryIPv4Address":"192.168.1.4",
        "ruleName":"[SECURITY-RULE-NAME]",
        "direction":"[DIRECTION-SPECIFIED-IN-RULE]",
        "type":"[ALLOW-OR-DENY-AS-SPECIFIED-IN-RULE]",
        "matchedConnections":125
        }
}
```

> **Note**
>
> The source IP address for the communication isn't logged. You can enable Virtual network flow logs for the NSG's virtual network, which logs all of the IP traffic flowing through a virtual network. Virtual network flow log data is written to an Azure Storage account in the same subscription and region of the virtual network. You can analyze the data with the traffic analytics capability of Azure Network Watcher.

View and analyze logs

If you send diagnostics data to:

- **Azure Monitor logs**: You can use the NSG analytics solution for enhanced insights. The solution provides visualizations for NSG rules that allow or deny traffic, per MAC address, of the network interface in a virtual machine.
- **Azure Storage account**: Data is written to a *PT1H.json* file. You can find the following logs:
  - Event log found in the path: *insights-logs-networksecuritygroupevent/resourceId=/SUBSCRIPTIONS/[ID]/RESOURCEGROUPS/[RESOURCE-GROUP-NAME-FOR-NSG]/PROVIDERS/MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/[NSG NAME]/y=[YEAR]/m=[MONTH/d=[DAY]/h=[HOUR]/m=[MINUTE]*
  - Rule counter log found in the path: *insights-logs-networksecuritygrouprulecounter/resourceId=/SUBSCRIPTIONS/[ID]/RESOURCEGROUPS*

/[RESOURCE-GROUP-NAME-FOR-NSG]/PROVIDERS/MICROSOFT.NETWORK/NETWORKSECURITYGROUPS/[NSG NAME]/y=[YEAR]/m=[MONTH/d=[DAY]/h=[HOUR]/m=[MINUTE]

To learn how to view resource log data, see Azure platform logs overview.

Related content

- For more information about activity logging, see Overview of Azure platform logs.

  Activity logging is enabled by default for NSGs created through either the ARM or classic deployment model. To determine which operations were completed on NSGs in the activity log, look for entries that contain the following resource types:
  - `Microsoft.ClassicNetwork/networkSecurityGroups`
  - `Microsoft.ClassicNetwork/networkSecurityGroups/securityRules`
  - `Microsoft.Network/networkSecurityGroups`
  - `Microsoft.Network/networkSecurityGroups/securityRules`

- To learn how to log diagnostic information for traffic flowing through the virtual network that match NSG rules, see how to manage virtual network flow logs.

Last updated on 03/02/2026

Test network latency between Azure VMs

Summarize this article for me

Summary

This article describes how to test network latency between Azure virtual machines (VMs) by using the publicly available tools Latte for Windows or SockPerf for Linux.

For the most accurate results, you should measure VM network latency with a tool that's designed for the task and excludes other types of latency, such as application latency. Latte and SockPerf provide the most relevant network latency results by focusing on Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) traffic. Most applications use these protocols, and this traffic has the largest effect on application performance.

Many other common network latency test tools, such as Ping, don't measure TCP or UDP traffic. Tools like Ping use Internet Control Message Protocol (ICMP), which applications don't use. ICMP traffic can be treated differently from application traffic and doesn't directly affect application performance. ICMP test results don't directly apply to workloads that use TCP and UDP.

Latte and SockPerf measure only TCP or UDP payload delivery times. These tools use the following approach to measure network latency between two physical or virtual computers:

1. Create a two-way communication channel between the computers by designating one as sender and one as receiver.
2. Send and receive packets in both directions and measure the round-trip time (RTT).

Tips and best practices to optimize network latency

To optimize VMs for network latency, observe the following recommendations when you create the VMs:

- Use the latest version of Windows or Linux.
- Enable Accelerated Networking for increased performance.
- Deploy VMs within an Azure proximity placement group.
- Create larger VMs for better performance.

Use the following best practices to test and analyze network latency:

1. As soon as you finish deploying, configuring, and optimizing network VMs, take baseline network latency measurements between deployed VMs to establish benchmarks.

2. Test the effects on network latency of changing any of the following components:

   - Operating system (OS) or network stack software, including configuration changes.
   - VM deployment methods, such as deploying to an availability zone or proximity placement group (PPG).
   - VM properties, such as Accelerated Networking or size changes.
   - Virtual network configuration, such as routing or filtering changes.

3. Always compare new test results to the baseline or to the latest test results before controlled changes.

4. Repeat tests whenever you observe or deploy changes.

Test VMs with Latte or SockPerf

Use the following procedures to install and test network latency with Latte for Windows or SockPerf for Linux.

Windows

Install Latte and configure VMs

1. Download the latest version of latte.exe to both VMs and put it in a separate folder such as *c:/tools*.

2. On the *receiver* VM, create a Windows Defender Firewall `allow` rule to allow the Latte traffic to arrive. It's easier to allow the *latte.exe* program by name than to allow specific inbound TCP ports. In the command, replace the `<path>` placeholder with the path you downloaded *latte.exe* to, such as *c:\tools*.

   Windows Command Prompt

   ```
   netsh advfirewall firewall add rule program=<path>latte.exe name="Latte"
   protocol=any dir=in action=allow enable=yes profile=ANY
   ```

Run Latte on the VMs

Run *latte.exe* from the Windows command line, not from PowerShell.

1. On the receiver VM, run the following command, replacing the `<receiver IP address>`, `<port>`, and `<iterations>` placeholders with your own values.

   Windows Command Prompt

   ```
   latte -a <receiver IP address>:<port> -i <iterations>
   ```

   - Around 65,000 iterations are enough to return representative results.
   - Any available port number is fine.

   The following example shows the command for a VM with an IP address of `10.0.0.4`:

   `latte -a 10.0.0.4:5005 -i 65100`

2. On the *sender* VM, run the same command as on the receiver, except with `-c` added to indicate the *client* or sender VM. Again, replace the `<receiver IP address>`, `<port>`, and `<iterations>` placeholders with your own values.

   Windows Command Prompt

   ```
   latte -c -a <receiver IP address>:<port> -i <iterations>
   ```

   For example:

   `latte -c -a 10.0.0.4:5005 -i 65100`

3. Wait for the results. Depending on how far apart the VMs are, the test could take a few minutes to finish. Consider starting with fewer iterations to test for success before running longer tests.

Next steps

- Reduce latency with an Azure proximity placement group.
- Optimize network throughput for Azure virtual machines.
- Allocate virtual machine network bandwidth.
- Test bandwidth and throughput.
- For more information about Azure virtual networking, see Azure Virtual Network FAQ.

Last updated on 03/02/2026

Test virtual machine network throughput by using NTTTCP

Summarize this article for me

Summary

This article describes how to use the free NTTTCP tool from Microsoft to test network bandwidth and throughput performance on Azure Windows or Linux virtual machines (VMs). A tool like NTTTCP targets the network for testing and minimizes the use of other resources that could affect performance.

Prerequisites

- An Azure account with an active subscription. Create an account for free.
- Two Windows or Linux virtual machines in Azure. Create a Windows virtual machine or create a Linux virtual machine.
  - To test throughput, you need two VMs of the same size to function as *sender* and *receiver*. The two VMs should be in the same proximity placement group or availability set, so you can use their internal IP addresses and exclude load balancers from the test.
  - Note the number of VM cores and the receiver VM IP address for the commands. Both the sender and receiver commands use the receiver's IP address.

> **Note**
>
> Testing by using a virtual IP is possible, but is beyond the scope of this article.

Examples used in this article

Expand table

Setting	Value
Receiver VM IP address	10.0.0.5
Number of VM cores	2

Test throughput with Windows VMs or Linux VMs

You can test throughput from Windows VMs by using NTTTCP or from Linux VMs by using NTTTCP-for-Linux.

Windows

Prepare VMs and install NTTTCP-for-Windows

1. On both the sender and receiver VMs, download the latest version of NTTTCP into a separate folder like **c:\tools**.

2. Open the Windows command line and navigate to the folder where you downloaded **ntttcp.exe**.

3. On the receiver VM, create a Windows Firewall `allow` rule to allow the NTTTCP traffic to arrive. It's easier to allow **nttcp.exe** by name than to allow specific inbound TCP ports. Run the following command, replacing `c:\tools` with your download path for **ntttcp.exe** if different.

   Windows Command Prompt

   ```
   netsh advfirewall firewall add rule program=c:\tools\ntttcp.exe
   name="ntttcp" protocol=any dir=in action=allow enable=yes profile=ANY
   ```

4. To confirm your configuration, use the following commands to test a single Transfer Control Protocol (TCP) stream for 10 seconds on the receiver and sender virtual machines:

   Receiver VM

   `ntttcp -r -m [<number of VM cores> x 2],*,<receiver IP address> -t 10 -P 1`

   Windows Command Prompt

   ```
   ntttcp -r -m 4,*,10.0.0.5 -t 10 -P 1
   ```

   Sender VM

   `ntttcp -s -m [<number of VM cores> x 2],*,<receiver IP address> -t 10 -P 1`

   Windows Command Prompt

   ```
   ntttcp -s -m 4,*,10.0.0.5 -t 10 -P 1
   ```

> **Note**
>
> Use the preceding commands only to test configuration.

> **Tip**
>
> When you run the test for the first time to verify setup, use a short test duration to get quick feedback. Once you verify the tool is working, extend the test duration to 300 seconds for the most accurate results.

Run throughput tests

Run the test for 300 seconds, or five minutes, on both the sender and receiver VMs. The sender and receiver must specify the same test duration for the `-t` parameter.

1. On the receiver VM, run the following command, replacing the `<number of VM cores>`, and `<receiver IP address>` placeholders with your own values.

   ```
   ntttcp -r -m [<number of VM cores> x 2],*,<receiver IP address> -t 300
   ```

   Windows Command Prompt

   ```
   ntttcp -r -m 4,*,10.0.0.5 -t 300
   ```

2. On the sender VM, run the following command. The sender and receiver commands differ only in the `-s` or `-r` parameter that designates the sender or receiver VM.

   ```
   ntttcp -s -m [<number of VM cores> x 2],*,<receiver IP address> -t 300
   ```

   Windows Command Prompt

   ```
   ntttcp -s -m 4,*,10.0.0.5 -t 300
   ```

3. Wait for the results.

When the test is complete, the output should be similar as the following example:

Output

```
C:\tools>ntttcp -s -m 4,*,10.0.0.5 -t 300
Copyright Version 5.39
Network activity progressing...
```

```
Thread  Time(s) Throughput(KB/s) Avg B / Compl
======  ======= ================ =============
     0  300.006        29617.328     65536.000
     1  300.006        29267.468     65536.000
     2  300.006        28978.834     65536.000
     3  300.006        29016.806     65536.000


#####  Totals:  #####


   Bytes(MEG)    realtime(s) Avg Frame Size Throughput(MB/s)
================ =========== ============== ================
    34243.000000     300.005       1417.829          114.141


Throughput(Buffers/s) Cycles/Byte       Buffers
===================== =========== =============
             1826.262       7.036    547888.000


DPCs(count/s) Pkts(num/DPC)   Intr(count/s) Pkts(num/intr)
============= ============= =============== ==============
     4218.744         1.708        6055.769          1.190


Packets Sent Packets Received Retransmits Errors Avg. CPU %
============ ================ =========== ====== ==========
    25324915          2161992       60412      0     15.075
```

Test throughput between a Windows VM and a Linux VM

To run NTTTCP throughput tests between a Windows VM and a Linux VM, enable no-sync mode by using the `-ns` flag on Windows or the `-N` flag on Linux.

Windows

To test with the Windows VM as the receiver, run the following command:

Windows Command Prompt

```
ntttcp -r -m [<number of VM cores> x 2],*,<Linux VM IP address> -t 300
```

To test with the Windows VM as the sender, run the following command:

Windows Command Prompt

```
ntttcp -s -m [<number of VM cores> x 2],*,<Linux VM IP address> -ns -t 300
```

Next steps

- Optimize network throughput for Azure virtual machines.
- Allocate virtual machine network bandwidth.
- Test VM network latency.
- For more information about Azure virtual networking, see Azure Virtual Network FAQ.

Last updated on 03/02/2026

Troubleshoot Azure VM connectivity problems

Summarize this article for me

Summary

This article helps administrators diagnose and resolve connectivity problems that affect Azure virtual machines (VMs).

Problems

- An Azure VM that’s deployed by using Resource Manager can't connect to another Azure VM in same virtual network.
- An Azure VM can't connect to the second network adapter of an Azure VM in same virtual network.
- An Azure VM can't connect to the internet.

To resolve these problems, follow the steps in the following section.

> **Note**
>
> You can use the following:
>
> - `netstat -an` to list the ports that the VM is listening to
> - Test-NetConnection module in PowerShell to display diagnostic information for a connection such as ping test and tcp test
> - Ensure to utilize tcping or other TCP-based testing tools, as ICMP traffic is deprioritized by many networking devices. Using TCP tests provides more consistent and reliable results, especially in Azure environments.

Resolution

Azure VM can't connect to another Azure VM in same virtual network

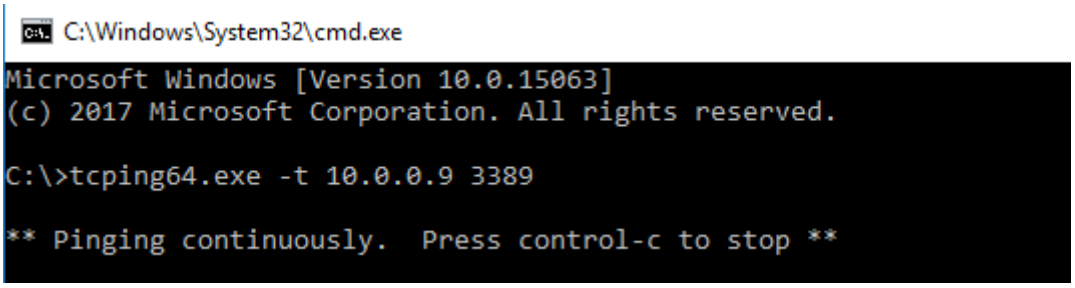
Step 1: Verify that VMs can communicate with each other.

1. Download TCping to your source VM.

2. Open a Command Prompt window.

3. Navigate to the folder in which you downloaded TCping.

4. Ping the destination from the source VM by using the following command:



Windows Command Prompt

```
tcping64.exe -t <destination VM address> 3389
```

> **Tip**
>
> If the ping test is successful, go to Step 3. Otherwise, go to the next step.

Step 2: Check the Network security group settings.

For each VM, check for default Inbound port rules of *Allow VNet Inbound* and *Allow Load Balancer Inbound*. Make sure to also check that there are no matching blocking rules that are listed below a lower-priority rule.

> **Note**
>
> Rules that have a lower number are matched first. For example, if you have a rule that has priority 1000 and 6500, the rule that has priority 1000 is matched first.

After that, try to ping the destination from the source VM again:

Windows Command Prompt

```
tcping64.exe -t <destination VM address> 3389
```

Step 3: Check whether you can connect to the destination VM by using Remote Desktop or SSH.

To connect by using Remote Desktop, follow these steps.

Windows:

1. Sign in to the Azure portal.
2. In the left menu, select **Virtual Machines**.
3. Select the virtual machine in the list.
4. On the page for the virtual machine, select **Connect**.

For more information, see How to connect and sign on to an Azure virtual machine running Windows.

Linux:

For more information, see Connect to a Linux VM in Azure.

If the Remote Desktop or SSH connection is successful, go to next step.

Step 4: Perform a connectivity check.

Run a connectivity check on the source VM, and check the response.

Windows: Check connectivity with Azure Network Watcher using PowerShell

Linux: Check connectivity with Azure Network Watcher using Azure CLI 2.0

The following is an example response:

```
ConnectionStatus : Unreachable
AvgLatencyInMs   :
MinLatencyInMs   :
MaxLatencyInMs   :
ProbesSent       : 100
ProbesFailed     : 100
Hops             : [
                     {
                       "Type": "Source",
                       "Id": "aaaaaaaa-0000-1111-2222-bbbbbbbbbbbb",
                       "Address": "10.1.1.4",
                       "ResourceId": "/subscriptions/aaaa0a0a-bb1b-cc2c-dd3d-
eeeeee4e4e4e/resourceGrou

ps/ContosoRG/providers/Microsoft.Network/networkInterfaces/appNic0/ipConfigurat
                    ions/ipconfig1",
```

```
                        "NextHopIds": [
                          "bbbbbbbb-1111-2222-3333-cccccccccccc"
                        ],
                        "Issues": []
                      },
                      {
                        "Type": "VirtualAppliance",
                        "Id": "bbbbbbbb-1111-2222-3333-cccccccccccc",
                        "Address": "10.1.2.4",
                        "ResourceId": "/subscriptions/aaaa0a0a-bb1b-cc2c-dd3d-
eeeeee4e4e4e/resourceGrou

ps/ContosoRG/providers/Microsoft.Network/networkInterfaces/fwNic/ipConfiguratio
                    ns/ipconfig1",
                        "NextHopIds": [
                          "cccccccc-2222-3333-4444-dddddddddddd"
                        ],
                        "Issues": []
                      },
```


Step 5: Fix the issue in the connectivity check result.

1. In the **Hops** section of the connectivity check response that you received, check the listed **issues**.

```
    },
    {
      "Type": "VirtualAppliance",
      "Id": "bbbbbbbb-1111-2222-3333-cccccccccccc",
      "Address": "10.1.3.4",
      "ResourceId": "/subscriptions/00000000-0000-0000-0000-00000
ps/ContosoRG/providers/Microsoft.Network/networkInterfaces/auNi
ns/ipconfig1",
      "NextHopIds": [
        "cccccccc-2222-3333-4444-dddddddddddd"
      ],
      "Issues": [
        {
          "Origin": "Outbound",
          "Severity": "Error",
          "Type": "NetworkSecurityRule",
          "Context": [
            {
              "key": "RuleName",
              "value": "UserRule_Port80"
            }
          ]
        }
      ]
    },
    {
      "Type": "VnetLocal",
      "Id": "eeeeeeee-4444-5555-6666-ffffffffffff",
      "Address": "10.1.4.4",
      "ResourceId": "/subscriptions/00000000-0000-0000-0000-00000
ps/ContosoRG/providers/Microsoft.Network/networkInterfaces/dbNi
```


2. Find the corresponding resolution in the following table, and follow the indicated steps to resolve the issues.

Expand table

Issue type	Value	Resolution action
NetworkSecurityRule	Name of the blocking NSG	You can delete the NSG rule or modify the rule as described here.
UserDefinedRoute	Name of the blocking UDR	If you don't require this route, delete the UDR. If you can't delete the route, update the route by using the appropriate address prefix and next hop. You can also adjust the Network Virtual Appliance to forward traffic appropriately. For more information, see: Virtual network traffic routing and Route network traffic with a route table using PowerShell.
CPU	Usage	Follow these recommendations that describe in Generic performance troubleshooting for Azure Virtual Machine

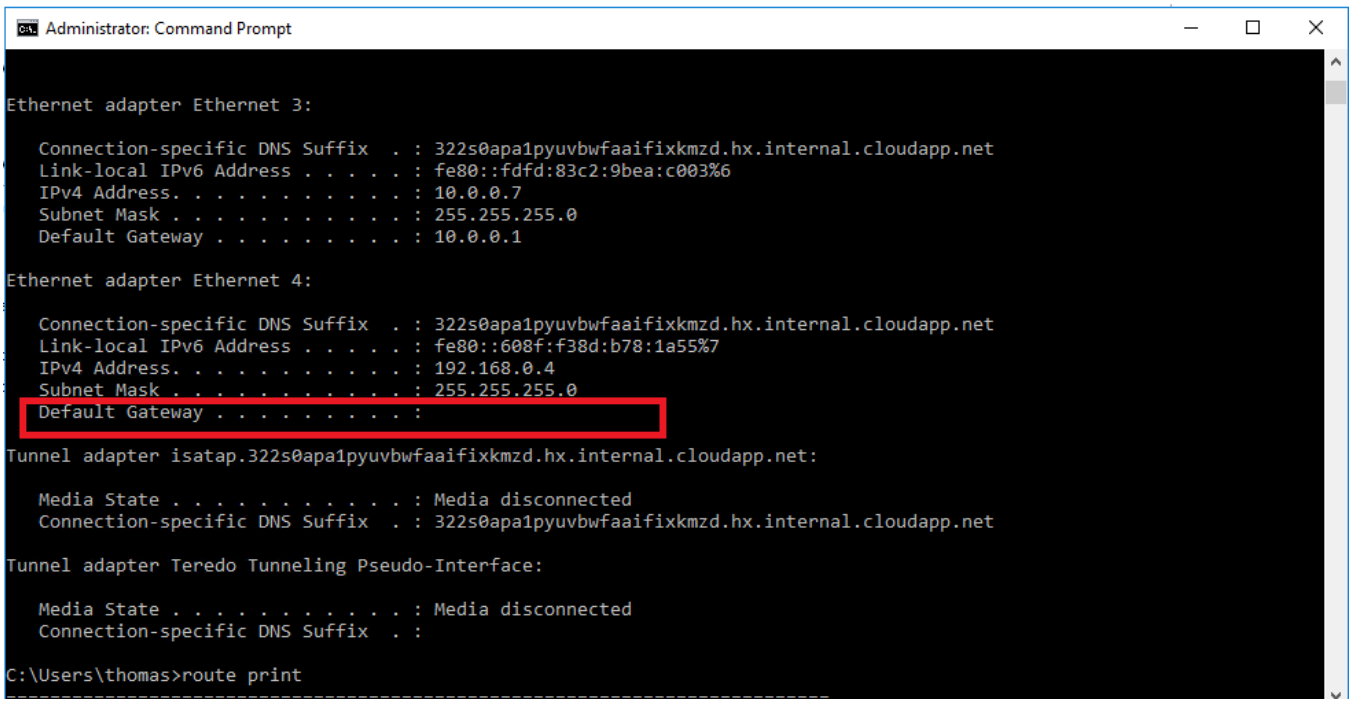
Issue type	Value	Resolution action
		running Linux or Windows .
Memory	Usage	Follow the recommendations that are described in Generic performance troubleshooting for Azure Virtual Machine running Linux or Windows .
Guest Firewall	Name of the firewall blocking	Follow these steps: Turn Windows Firewall on or off .
DNS Resolution	Name of the DNS	Follow these steps: Azure DNS troubleshooting guide and Name resolution for resources in Azure virtual networks.
Socket Error	Not applicable	The specified port is already in use by another application. Try to use a different port.

3. Run the connectivity check again to determine whether the problem is resolved.

Azure VM can't connect to the second network adapter of an Azure VM in same virtual network

Step 1: Make sure that the second network adapter is enabled to talk outside the subnet.

By default, secondary network adapters (also known as network interface cards, or network adapters) aren't configured to have a default gateway. Therefore, the traffic flow on the secondary adapter will be limited to the same subnet.



If users want to enable secondary network adapters to talk outside their own subnet, they must add an entry to the routing table to configure the gateway. To do this, follow these steps:

1. On the VM that has the second network adapter configured, open a Command Prompt window as an administrator.

2. Run the following command to add the entry in routing table:

   Windows Command Prompt

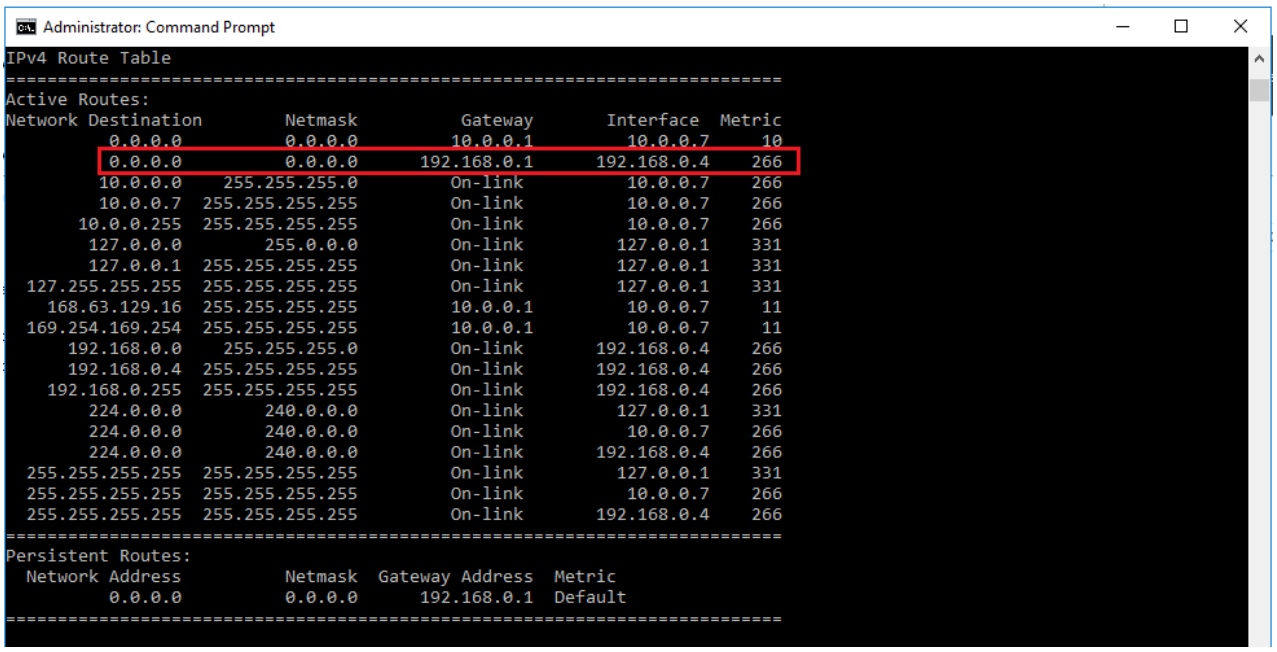
   ```
   Route add 0.0.0.0 mask 0.0.0.0 -p <Gateway IP>
   ```

   For example, if the second IP address is 192.168.0.4, the gateway IP should be 192.168.0.1. You have to run the following command:

   Windows Command Prompt

   ```
   Route add 0.0.0.0 mask 0.0.0.0 -p 192.168.0.1
   ```

3. Run route print. If the entry is added successfully, you'll see an entry that resembles the following:

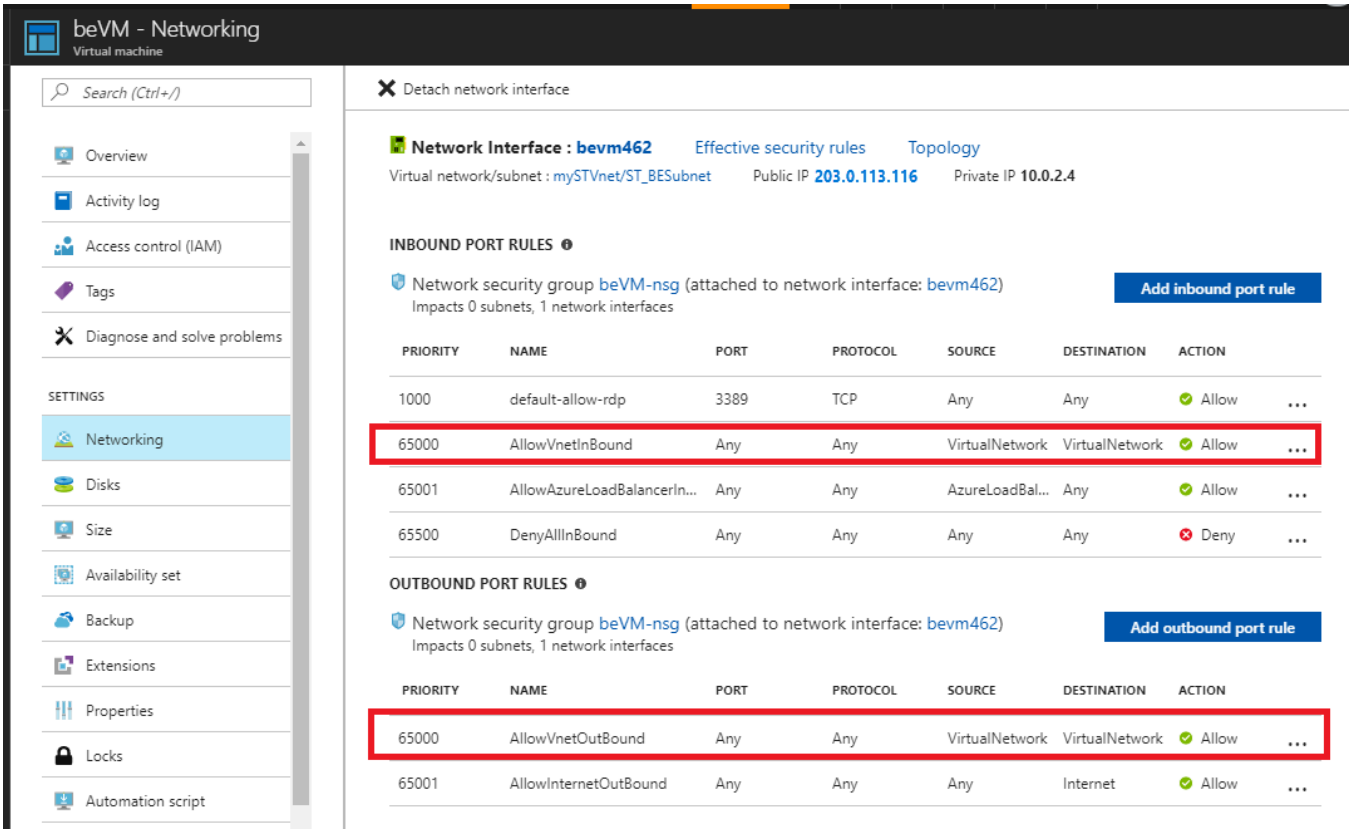


Now, try to connect to secondary network adapter. If the connection is still unsuccessful, go to next step.

Step 2: Check NSG settings for the network adapters.

For both the primary and secondary network adapters, check the default Inbound port rules of **Allow VNet Inbound** and **Allow Load Balancer** allow inbound on both network adapters. You

should also make sure that there are no matching blocking rules that have a lower-priority rule above them.



Step 3: Run a connectivity check to the secondary network adapter.

1. Run a connectivity check to the secondary network adapter.
2. Run a connectivity check across the environment to make sure that the process works end to end.

For more information about how to run the connectivity check, see the following articles:

Windows: Check connectivity with Azure Network Watcher using PowerShell

Linux: Check connectivity with Azure Network Watcher using Azure CLI 2.0

The following is an example response:

```
ConnectionStatus : Unreachable
AvgLatencyInMs   :
MinLatencyInMs   :
MaxLatencyInMs   :
ProbesSent       : 100
ProbesFailed     : 100
Hops             : [
                     {
                       "Type": "Source",
```

```
                        "Id": "aaaaaaaa-0000-1111-2222-bbbbbbbbbbbb",
                        "Address": "10.1.1.4",
                        "ResourceId": "/subscriptions/aaaa0a0a-bb1b-cc2c-dd3d-
eeeeee4e4e4e/resourceGrou

ps/ContosoRG/providers/Microsoft.Network/networkInterfaces/appNic0/ipConfigurat
                    ions/ipconfig1",
                        "NextHopIds": [
                          "bbbbbbbb-1111-2222-3333-cccccccccccc"
                        ],
                        "Issues": []
                      },
                      {
                        "Type": "VirtualAppliance",
                        "Id": "bbbbbbbb-1111-2222-3333-cccccccccccc",
                        "Address": "10.1.2.4",
                        "ResourceId": "/subscriptions/aaaa0a0a-bb1b-cc2c-dd3d-
eeeeee4e4e4e/resourceGrou

ps/ContosoRG/providers/Microsoft.Network/networkInterfaces/fwNic/ipConfiguratio
                    ns/ipconfig1",
                        "NextHopIds": [
                          "cccccccc-2222-3333-4444-dddddddddddd"
                        ],
                        "Issues": []
                      },
```


Step 4: Refer the table under Step 5, and follow these steps to resolve the issues.

Azure VM can't connect to the internet

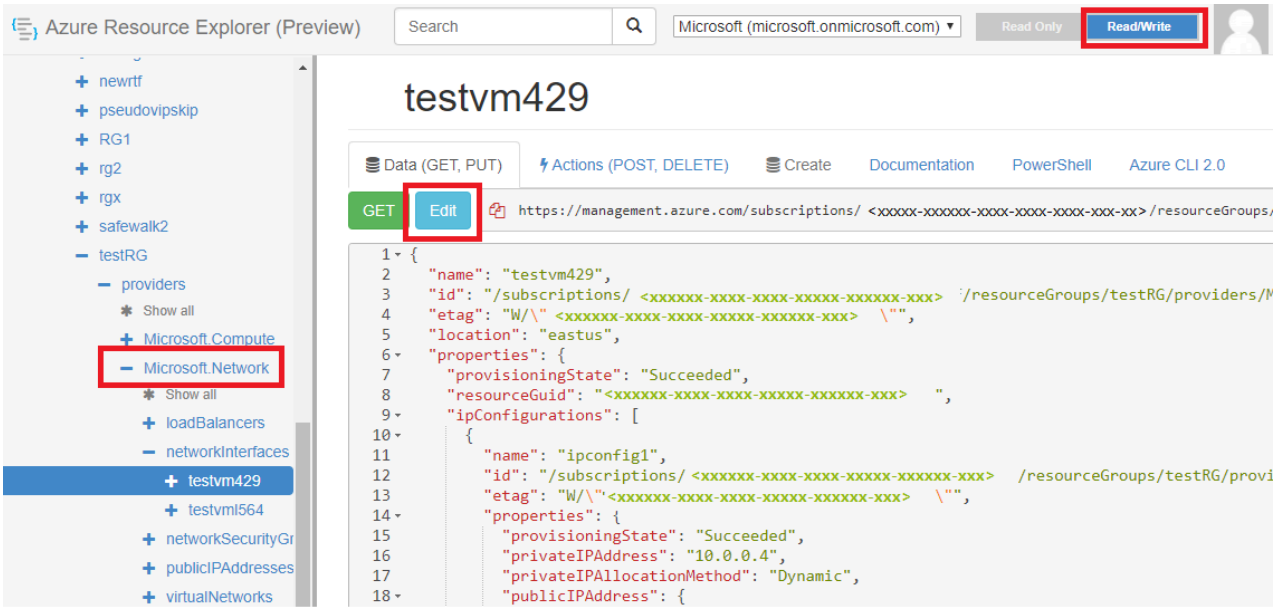
Step 1: Check whether the network adapter is in a failed state.

Follow these steps to check the state of the NIC:

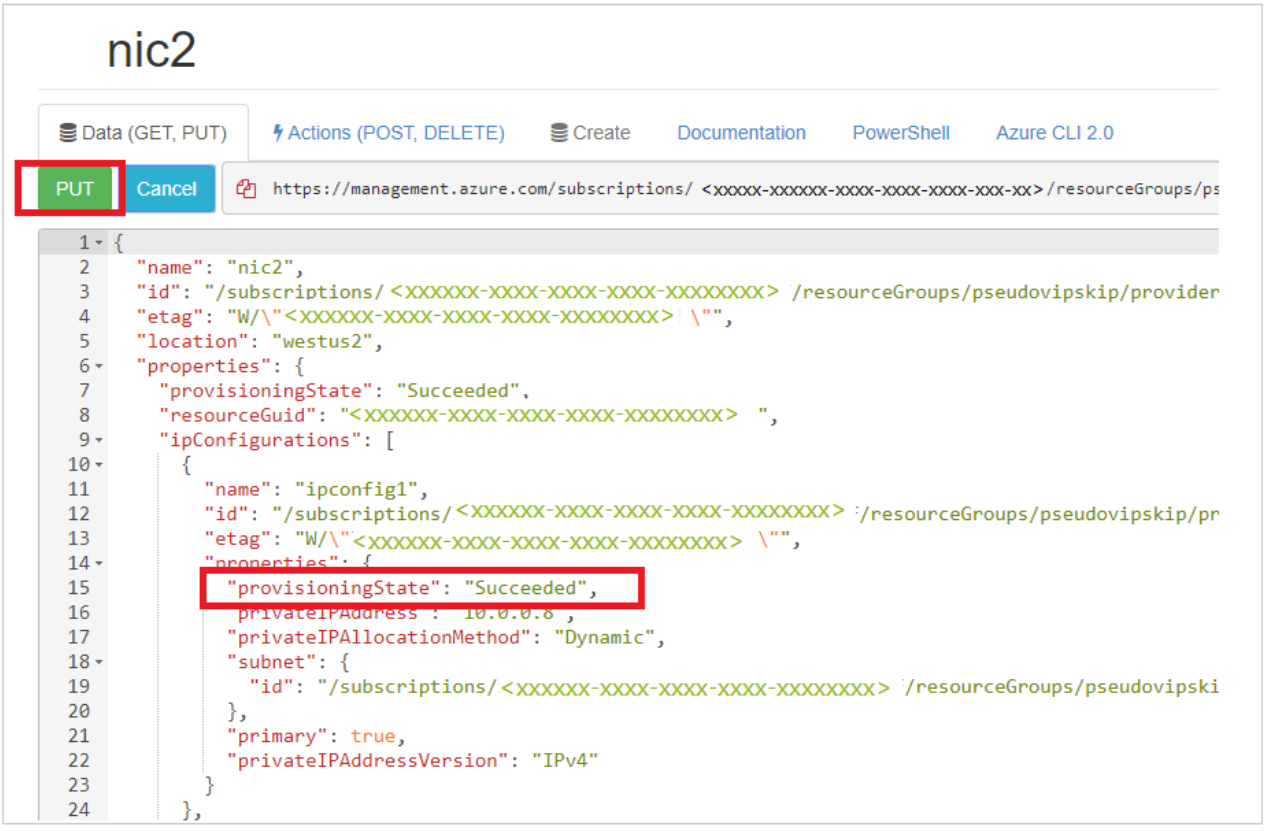
1. Log in to the Resource Explorer portal.
2. In the left pane, select **Subscriptions**.
3. In the left pane, select the resource group that the affected network adapter or VM belongs to.
4. Go to the **Microsoft Network**.
5. Select the **Network Interfaces** option.
6. Select the affected network interface.

7. Select the **Read/Write** option at the top of the portal.

8. Select the **Edit** option.



> ⓘ **Note**
>
> After you select the **Edit** option, the "Get" option changes to a **Put** option.



9. Select the affected network interface, and then select the **Put** option.

> ⓘ **Note**

After you make this selection, the **ProvisioningState** is displayed as **Updating**. The same result is shown on the regular Azure Resource Manager portal. If the operation is completed successfully, the **ProvisioningState** value changes to **Succeeded**, as shown.

10. Refresh your portal. The network adapter should be in a success state.

Step 2: Follow Step 4 to run a connectivity check.

Step 3: Refer the table under Step 5, and follow the steps to resolve the issues.

Next steps

Troubleshooting connectivity problems between Azure VMs

Last updated on 03/02/2026

Troubleshoot outbound SMTP connectivity problems in Azure

Summarize this article for me

Summary

Troubleshooting outbound SMTP connectivity in Azure can be challenging when email messages fail to reach external domains. Outbound email messages that are sent directly to external domains (such as outlook.com and gmail.com) from a virtual machine (VM) are made available only to certain subscription types in Microsoft Azure. This article helps you resolve SMTP connectivity issues and configure reliable email delivery from your Azure VMs.

> **Important**
>
> For the following examples, the process applies mainly to Azure Virtual Machines & Azure Virtual Machine Scale Sets resources (`Microsoft.Compute/virtualMachines` & `Microsoft.Compute/virtualMachineScaleSets`).
>
> It's possible to use port 25 for outbound communication on Azure App Service and Azure Functions through the virtual network integration feature or when App Service Environment v3 is used.
>
> It's also possible to send port 25 outbound communication through Azure Firewall. However, the following subscription limitations described still apply. Sending email on Port 25 is unsupported for all other Azure Platform-as-a-Service (PaaS) resources.

Recommended method of sending email

We recommend you use authenticated SMTP relay services to send email from Azure VMs or from Azure App Service. Connections to authenticated SMTP relay services are typically on TCP port 587 and isn't blocked. These services are used in part to maintain IP reputation that is critical for delivery reliability. Azure Communication Services offers an authenticated SMTP relay service. Ensure that the default rate limits are appropriate for your application and open a support case to raise them if needed.

Using these email delivery services on authenticated SMTP port 587 isn't restricted in Azure, regardless of the subscription type.

Enterprise and MCA-E

For VMs and Azure Firewall that are deployed in standard Enterprise Agreement or Microsoft Customer Agreement for enterprise (MCA-E) subscriptions, the outbound SMTP connections on TCP port 25 aren't blocked. However, there's no guarantee that external domains accept the incoming emails from the VMs and Azure Firewall. For emails rejected or filtered by the external domains, contact the email service providers of the external domains to resolve the problems. These problems aren't covered by Azure support.

For Enterprise Dev/Test subscriptions, port 25 is blocked by default. It's possible to have this block removed. To request to have the block removed, go to the **Cannot send email (SMTP-Port 25)** section of the **Diagnose and Solve** section in the Azure Virtual Network resource in the Azure portal and run the diagnostic. This process exempts the qualified enterprise dev/test subscriptions automatically.

After the subscription is exempted from this block, the VMs must be stopped, deallocated, and then restarted to get the new network policy, all VMs in that subscription are exempted going forward. If the virtual network owned by the exempted subscription has a delegated subnet (to an App Service Environment for example), you must add and remove a new temporary subnet in the Virtual Network. The exemption applies only to the subscription requested and only to VM traffic that is routed directly to the internet.

All other subscription types

The Azure platform blocks outbound SMTP connections on TCP port 25 for deployed VMs. This block is to ensure better security for Microsoft partners and customers, protect Microsoft's Azure platform, and conform to industry standards.

If you're using a subscription type that isn't an Enterprise Agreement or MCA-E, we encourage you to use an authenticated SMTP relay service, as outlined earlier in this article.

Changing subscription type

If you change your subscription type from Enterprise Agreement or MCA-E to another type of subscription, changes to your deployments might result in outbound SMTP being blocked.

Need help? Contact support

If you're using an Enterprise Agreement or MCA-E subscription and still need help, contact support to get your problem resolved quickly. Use this issue type: **Technical** > **Virtual Network** > **Cannot send email (SMTP/Port 25)**.

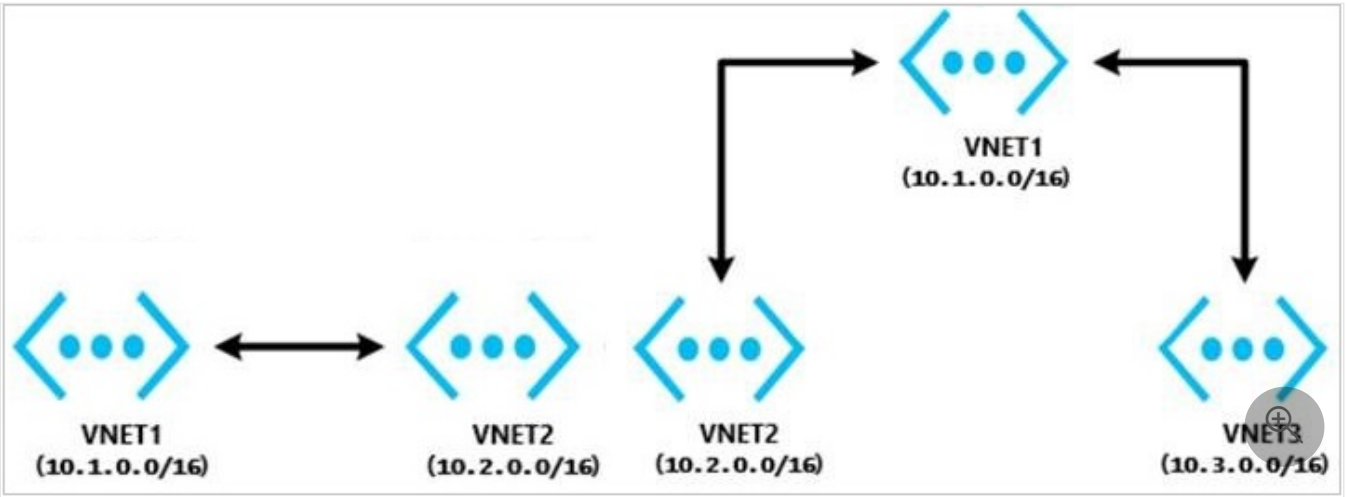
Last updated on 03/02/2026

Troubleshoot virtual network peering issues

Summarize this article for me

Summary

This troubleshooting guide provides steps to help you resolve most virtual network peering issues.



Configure virtual network peering between two virtual networks

Are the virtual networks in the same subscription or in different subscriptions?

The virtual networks are in the same subscription

To configure virtual network peering for the virtual networks that are in the same subscription, use the methods in the following articles:

- If the virtual networks are in the *same region*, see Create a peering.
- If the virtual networks are in the *different regions*, see Virtual network peering.

> **Note**
>
> Connectivity doesn't work over global virtual network peering for the following resources:
>
> - Virtual machines (VMs) behind Basic internal load balancer (ILB) SKU
> - Redis cache (uses Basic ILB SKU)

- Application gateway v1 (uses Basic ILB SKU)
- Virtual machine scale sets (uses Basic ILB SKU)
- Azure Service Fabric clusters (uses Basic ILB SKU)
- SQL Server Always On (uses Basic ILB SKU)
- Azure App Service Environment for Power Apps (uses Basic ILB SKU)
- Azure API Management (uses Basic ILB SKU)
- Microsoft Entra Domain Services (uses Basic ILB SKU)

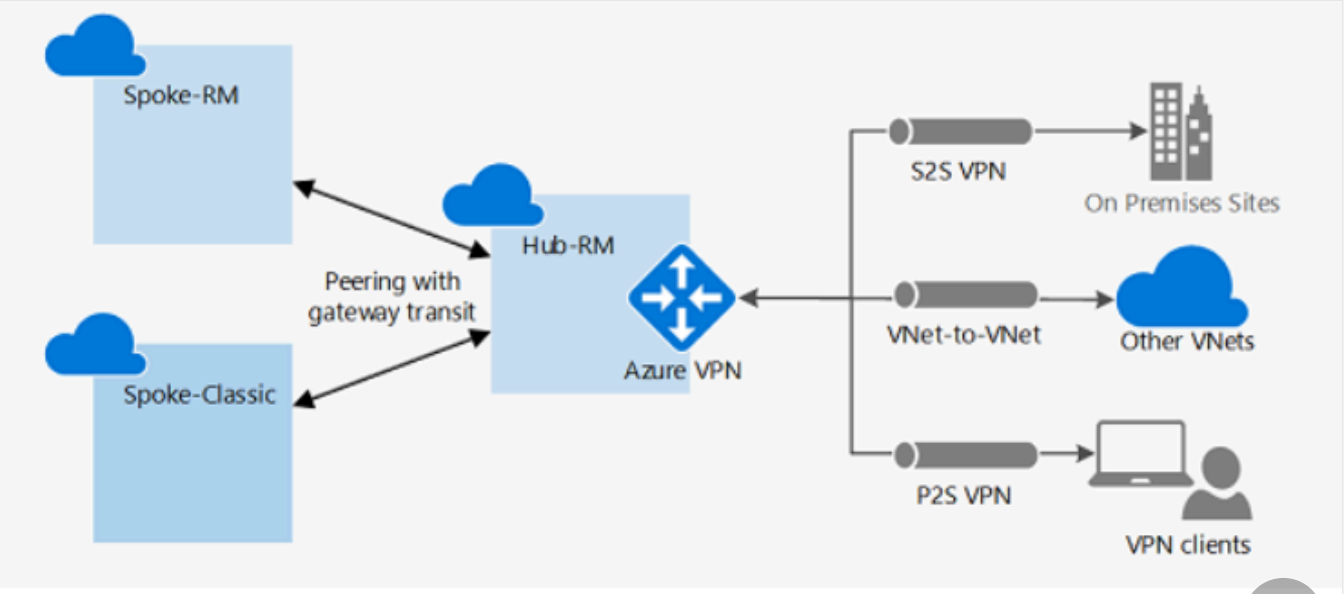
For more information, see the requirements and constraints of global peering.

The virtual networks are in different subscriptions or Active Directory tenants

To configure virtual network peering for virtual networks in different subscriptions or Active Directory tenants, see Create a virtual network peering between different subscriptions.

> **Note**
>
> To configure network peering, you must have **Network Contributor** permissions in both subscriptions. For more information, see Peering permissions.

Configure virtual network peering with hub-spoke topology that uses on-premises resources



On-Premises Spoke

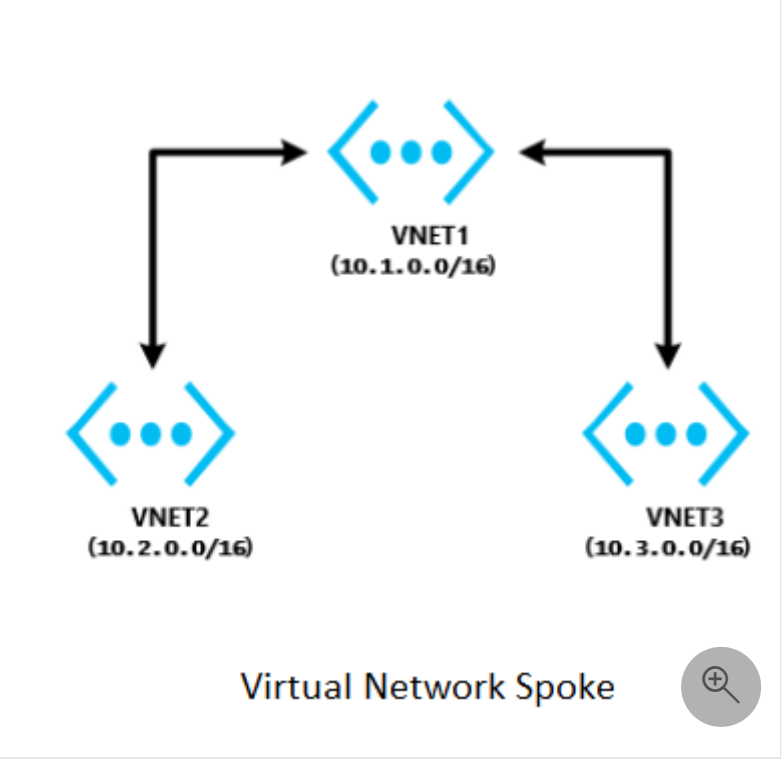
For a site-to-site connection or an ExpressRoute connection

Follow the steps in: Configure VPN gateway transit for virtual network peering.

For point-to-site connections

1. Follow the steps in: Configure VPN gateway transit for virtual network peering.
2. After virtual network peering is established or changed, download and reinstall the point-to-site package so that the point-to-site clients get the updated routes to the spoke virtual network.

Configure virtual network peering with hub-spoke topology virtual network



The virtual networks are in the same region

1. In the hub virtual network, configure a network virtual appliance (NVA).
2. In the spoke virtual networks, have user-defined routes with the next hop type "network virtual appliance" applied.

For more information, see Service chaining.

> **Note**

If you require help to set up an NVA, contact the NVA vendor.

For help with troubleshooting the NVA device setup and routing, see Network virtual appliance issues in Azure.

The virtual networks are in different regions

Transit over global virtual network peering is now supported. Connectivity does not work over global virtual network peering for the following resources:

- VMs behind Basic ILB SKU
- Redis cache (uses Basic ILB SKU)
- Application gateway (uses Basic ILB SKU)
- Scale sets (uses Basic ILB SKU)
- Service Fabric clusters (uses Basic ILB SKU)
- SQL Server Always On (uses Basic ILB SKU)
- App Service Environment (uses Basic ILB SKU)
- API Management (uses Basic ILB SKU)
- Microsoft Entra Domain Services (uses Basic ILB SKU)

To learn more about global peering requirements and restraints, see Virtual network peering.

Troubleshoot a connectivity issue between two peered virtual networks

Sign in to the Azure portal with an account that has the necessary roles and permissions. Select the virtual network, select **Peering**, and then check the **Status** field. What is the status?

The peering status is "Connected"

To troubleshoot this issue:

1. Check the network traffic flows:

   Use Connection Troubleshoot and IP flow verify from the source VM to the destination VM to determine whether there is an NSG or UDR that is causing interference in traffic flows.

   If you're using a firewall or NVA:

   a. Document the UDR parameters so that you can restore them after this step is complete.

b. Remove the UDR from the source VM subnet or NIC that points to the NVA as the next hop. Verify connectivity from the source VM directly to the destination that is bypassing the NVA. If this step doesn't work, see the NVA troubleshooter.

2. Take a network trace:

   a. Start a network trace on the destination VM. For Windows, you can use **Netsh**. For Linux, use **TCPDump**.

   b. Run **TcpPing** or **PsPing** from the source to the destination IP.

      This is an example of a **TcpPing** command: `tcping64.exe -t <destination VM address> 3389`

   c. After the **TcpPing** is complete, stop the network trace on the destination.

   d. If packets arrive from the source, there is no networking issue. Examine both the VM firewall and the application listening on that port to locate the configuration issue.

> **Note**
>
> You can't connect to the following resource types over global virtual network peering (virtual networks in different regions):
>
> - VMs behind Basic ILB SKU
> - Redis cache (uses Basic ILB SKU)
> - Application gateway (uses Basic ILB SKU)
> - Scale sets (uses Basic ILB SKU)
> - Service Fabric clusters (uses Basic ILB SKU)
> - SQL Server Always On (uses Basic ILB SKU)
> - App Service Environment (uses Basic ILB SKU)
> - API Management (uses Basic ILB SKU)
> - Microsoft Entra Domain Services (uses Basic ILB SKU)

For more information, see the requirements and constraints of global peering.

The peering status is "Disconnected"

To resolve this issue, delete the peering from both virtual networks, and then re-create them.

Troubleshoot a connectivity issue between a hub-spoke virtual network and an on-premises resource

Does your network use a third-party NVA or VPN gateway?

My network uses a third-party NVA or VPN gateway

To troubleshoot connectivity issues that affect a third-party NVA or VPN gateway, see the following articles:

- NVA troubleshooter
- Service chaining

My network does not use a third-party NVA or VPN gateway

Do the hub virtual network and the spoke virtual network have a VPN gateway?

Both the hub virtual network and the spoke virtual network have a VPN gateway

Using a remote gateway isn't supported.

If the spoke virtual network already has a VPN gateway, the **Use remote gateway** option isn't supported on the spoke virtual network. This is because of a virtual network peering limitation.

Both the hub virtual network and the spoke virtual network do not have a VPN gateway

For site-to-site or Azure ExpressRoute connections, check the following primary causes of connectivity issues to the remote virtual network from on-premises:

- On the virtual network that has a gateway, verify that the **Allow forwarded traffic** check box is selected.
- On the virtual network that doesn't have a gateway, verify that the **Use remote gateway** check box is selected.
- Have your network administrator check your on-premises devices to verify that they all have the remote virtual network address space added.

For point-to-site connections:

- On the virtual network that has a gateway, verify that the **Allow forwarded traffic** check box is selected.
- On the virtual network that doesn't have a gateway, verify that the **Use remote gateway** check box is selected.
- Download and reinstall the point-to-site client package. Virtual network routes that are newly peered don't automatically add routes to point-to-site clients.

Troubleshoot a hub-spoke network connectivity issue between spoke virtual networks in the same region

A hub network must include an NVA. Configure UDRs in spokes that have an NVA set as the next hop, and enable **Allow forwarded traffic** in the hub virtual network.

For more information, see Service chaining, and discuss these requirements with the NVA vendor of your choice.

Troubleshoot a hub-spoke network connectivity issue between spoke virtual networks in different regions

Transit over global virtual network peering is now supported. Connectivity doesn't work over global virtual network peering for the following resources:

- VMs behind Basic ILB SKU
- Redis cache (uses Basic ILB SKU)
- Application gateway (uses Basic ILB SKU)
- Scale sets (uses Basic ILB SKU)
- Service Fabric clusters (uses Basic ILB SKU)
- SQL Server Always On (uses Basic ILB SKU)
- App Service Environment (uses Basic ILB SKU)
- API Management (uses Basic ILB SKU)
- Microsoft Entra Domain Services (uses Basic ILB SKU)

For more information, see the requirements and constraints of global peering and Different VPN Topologies.

Troubleshoot a hub-spoke network connectivity issue between a web app and the spoke virtual network

To troubleshoot this issue:

1. Sign in to the Azure portal.
2. In the web app, select **networking**, and then select **VNet Integration**.
3. Check whether you can see the remote virtual network. Manually enter the remote virtual network address space (**Sync Network** and **Add Routes**).

For more information, see the following articles:

- Integrate your app with an Azure virtual network
- About Point-to-Site VPN routing

Troubleshoot a virtual network peering configuration error message

Current tenant `<TENANT ID>` isn't authorized to access linked subscription

To resolve this issue, see Create a virtual network peering between different subscriptions.

Not connected

To resolve this issue, delete the peering from both virtual networks, and then recreate them.

Failed to peer a Databricks virtual network

To resolve this issue, configure the virtual network peering under **Azure Databricks**, and then specify the target virtual network by using **Resource ID**. For more information, see Peer a Databricks virtual network to a remote virtual network.

The remote virtual network lacks a gateway

This issue occurs when you peer virtual networks from different tenants and later want to configure `Use Remote Gateways`. A limitation of the Azure portal is that it can't validate the presence of a virtual network gateway in another tenant's virtual network.

There are two ways to resolve the issue:

- Delete the peerings and activate the `Use Remote Gateways` option when you create a new peering.
- Use PowerShell or CLI, instead of the Azure portal, to enable `Use Remote Gateways`.

Next steps

- Troubleshooting connectivity problems between Azure VMs

Last updated on 03/02/2026

Troubleshooting connectivity problems between Azure VMs

Summarize this article for me

Summary

You might experience connectivity problems between Azure virtual machines (VMs). This article provides troubleshooting steps to help you resolve this problem.

If your Azure issue is not addressed in this article, visit the Azure forums on Microsoft Q & A and Stack Overflow. You can post your issue in these forums, or post to @AzureSupport on Twitter. You also can submit an Azure support request. To submit a support request, on the Azure support page, select **Get support**.

Symptom

One Azure VM can't connect to another Azure VM.

Troubleshooting guidance

1. Check whether NIC is misconfigured
2. Check whether network traffic is blocked by NSG or UDR
3. Check whether network traffic is blocked by VM firewall
4. Check whether VM app or service is listening on the port
5. Check whether the problem is caused by SNAT
6. Check whether traffic is blocked by ACLs for the classic VM
7. Check whether the endpoint is created for the classic VM
8. Try to connect to a VM network share
9. Check Inter-VNet connectivity

> **Note**
>
> You can also use Test-NetConnection module in PowerShell to diagnose information for a connection.

Troubleshooting steps

Follow these steps to troubleshoot the problem. After you complete each step, check whether the problem is resolved.

Step 1: Check whether NIC is misconfigured

Follow the steps in How to reset network interface for Azure Windows VM.

If the problem occurs after you modify the network interface (NIC), follow these steps:

Multi-NIC VMs

1. Add a NIC.
2. Fix the problems in the bad NIC or remove the bad NIC. Then add the NIC again.

For more information, see Add network interfaces to or remove from virtual machines.

Single-NIC VM

- Redeploy Windows VM
- Redeploy Linux VM

Step 2: Check whether network traffic is blocked by NSG or UDR

Use Network Watcher IP Flow Verify and Connection troubleshoot to determine whether there's a Network Security Group (NSG) or User-Defined Route (UDR) that is interfering with traffic flow. You may need to add inbound rules on both NSGs. The rules must be at the subnet level and the virtual machine's interface level.

Step 3: Check whether network traffic is blocked by VM firewall

Disable the firewall, and then test the result. If the problem is resolved, verify the firewall settings, and then re-enable the firewall.

Step 4: Check whether VM app or service is listening on the port

You can use one of the following methods to check whether the VM app or service is listening on the port.

- Run the following commands to check whether the server is listening on that port.

Windows VM

```
Console
netstat -ano
```

Linux VM

```
Console
netstat -l
```

- Run the **telnet** command on the virtual machine itself to test the port. If the test fails, the application or service isn't configured to listen on that port.

Step 5: Check whether the problem is caused by SNAT

In some scenarios, the VM is placed behind a load balance solution that has a dependency on resources outside of Azure. In these scenarios, if you experience intermittent connection problems, the problem may be caused by SNAT port exhaustion. To resolve the issue, create a VIP (or ILPIP for classic) for each VM that is behind the load balancer and secure with NSG or ACL.

Step 6: Check whether traffic is blocked by ACLs for the classic VM

An access control list (ACL) provides the ability to selectively permit or deny traffic for a virtual machine endpoint. For more information, see Manage the ACL on an endpoint.

Step 7: Check whether the endpoint is created for the classic VM

All VMs that you create in Azure by using the classic deployment model can automatically communicate over a private network channel with other virtual machines in the same cloud service or virtual network. However, computers on other virtual networks require endpoints to direct the inbound network traffic to a virtual machine. For more information, see How to set up endpoints.

Step 8: Try to connect to a VM network share

If you can't connect to a VM network share, the problem may be caused by unavailable NICs in the VM. To delete the unavailable NICs, see How to delete the unavailable NICs

Step 9: Check Inter-VNet connectivity

Use Network Watcher IP Flow Verify and NSG Flow Logging to determine whether there's an NSG or UDR that is interfering with traffic flow. You can also verify your Inter-VNet configuration here.

Need help? Contact support

If you still need help, contact support to get your issue resolved quickly.

Last updated on 03/02/2026

Troubleshooting: Failed to delete a virtual network in Azure

Summarize this article for me

Summary

You might receive errors when you try to delete a virtual network in Microsoft Azure. This article provides troubleshooting steps to help you resolve this problem.

If your Azure issue is not addressed in this article, visit the Azure forums on Microsoft Q & A and Stack Overflow. You can post your issue in these forums, or post to @AzureSupport on Twitter. You also can submit an Azure support request. To submit a support request, on the Azure support page, select **Get support**.

Troubleshooting guidance

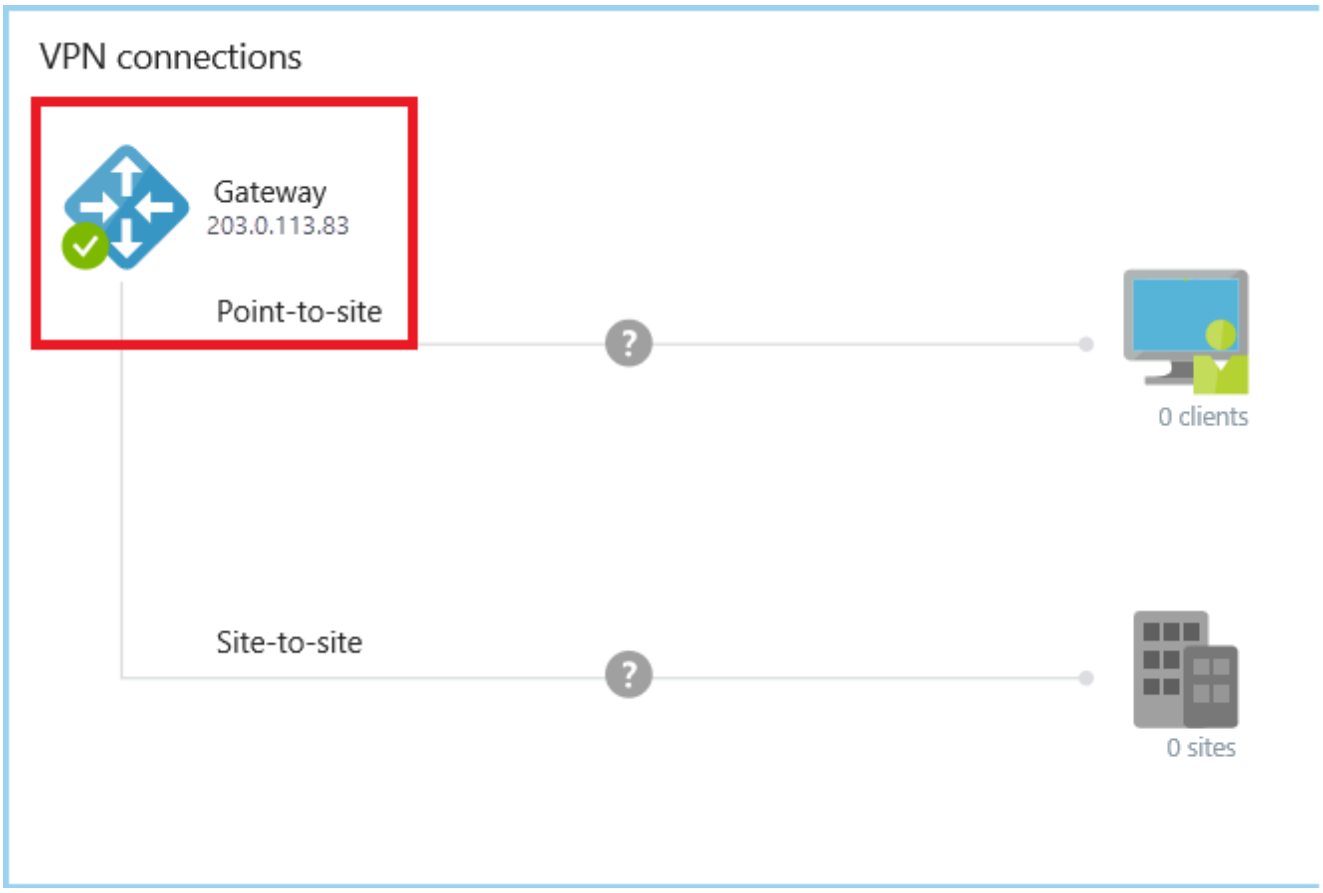
1. Check whether a virtual network gateway is running in the virtual network.
2. Check whether an application gateway is running in the virtual network.
3. Check whether Azure container instances still exist in the virtual network.
4. Check whether Microsoft Entra Domain Service is enabled in the virtual network.
5. Check whether the virtual network is connected to other resources.
6. Check whether a virtual machine is still running in the virtual network.
7. Check whether the virtual network is stuck in migration.
8. Check whether the virtual network was used by a web app for virtual network integration.

Troubleshooting steps

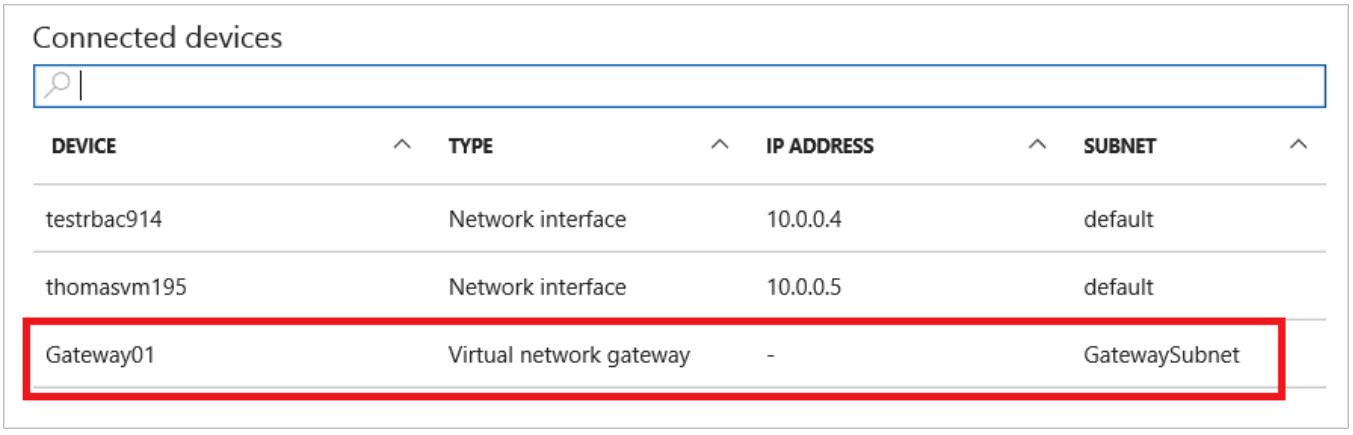
Check whether a virtual network gateway is running in the virtual network

To remove the virtual network, you must first remove the virtual network gateway.

For classic virtual networks, go to the **Overview** page of the classic virtual network in the Azure portal. In the **VPN connections** section, if the gateway is running in the virtual network, the IP address of the gateway is shown.



For virtual networks, go to the **Overview** page of the virtual network. Check **Connected devices** for the virtual network gateway.



DEVICE	TYPE	IP ADDRESS	SUBNET
testrbac914	Network interface	10.0.0.4	default
thomasvm195	Network interface	10.0.0.5	default
Gateway01	Virtual network gateway	-	GatewaySubnet

Before you can remove the gateway, first remove any **Connection** objects in the gateway.

Check whether an application gateway is running in the virtual network

Go to the **Overview** page of the virtual network. Check the **Connected devices** for the application gateway.

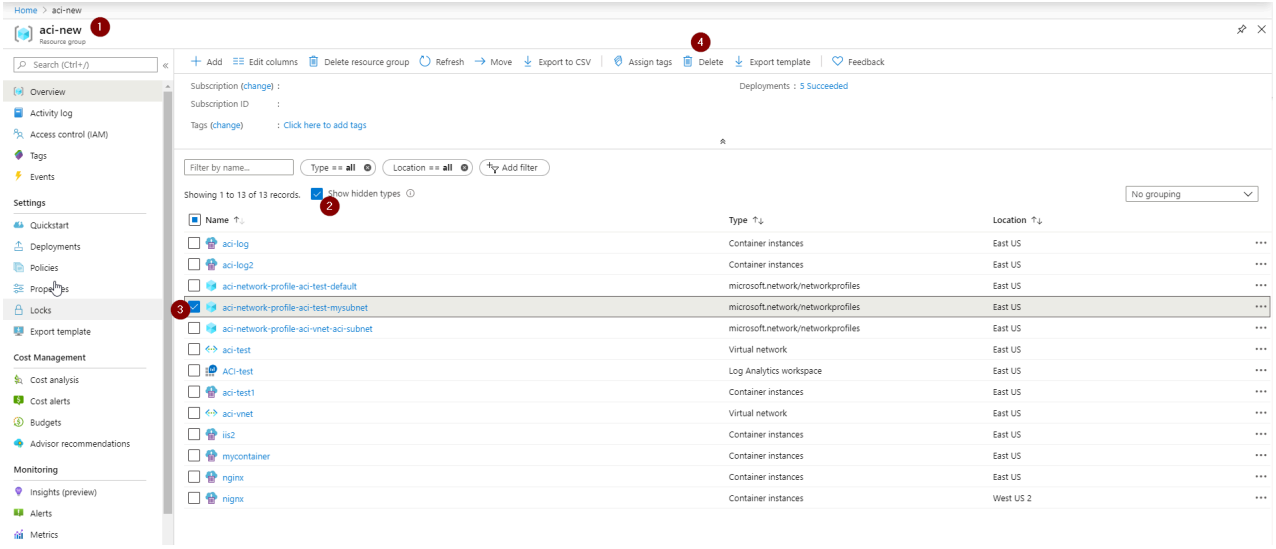
Connected devices

DEVICE	TYPE	IP ADDRESS	SUBNET
testrbac914	Network interface	10.0.0.4	default
thomasvm195	Network interface	10.0.0.5	default
AppGateway-Test	Application gateway		gateway

If there's an application gateway, you must remove it before you can delete the virtual network.

Check whether Azure container instances still exist in the virtual network

1. In the Azure portal, go to the resource group's **Overview** page.
2. In the header for the list of the resource group's resources, select **Show hidden types**. The network profile type is hidden in the Azure portal by default.
3. Select the network profile related to the container groups.
4. Select **Delete**.



5. Delete the subnet or virtual network again.

If these steps don't resolve the issue, use these Azure CLI commands to clean up resources.

Check whether Microsoft Entra Domain Service is enabled in the virtual network

If the Active Directory Domain Service is enabled and connected to the virtual network, you can't delete this virtual network.

To disable the service, see Disable Microsoft Entra Domain Services using the Azure portal.

Check whether the virtual network is connected to other resources

Check for Circuit Links, connections, and virtual network peerings. Any of these can cause a virtual network deletion to fail.

The recommended deletion order is as follows:

1. Gateway connections
2. Gateways
3. IPs
4. Virtual network peerings
5. App Service Environment (ASE)

Check whether a virtual machine is still running in the virtual network

Make sure that no virtual machine is in the virtual network.

Check whether the virtual network is stuck in migration

If the virtual network is stuck in a migration state, it can't be deleted. Run the following command to abort the migration, and then delete the virtual network.

Azure PowerShell

```
Move-AzureVirtualNetwork -VirtualNetworkName "Name" -Abort
```

Check whether the virtual network was used by a web app for virtual network integration

If the virtual network was integrated with a web app in the past, then the web app was deleted without disconnecting the virtual network integration, see Deleting the App Service plan or web app before disconnecting the virtual network integration.

Next steps

- Azure Virtual Network
- Azure Virtual Network frequently asked questions (FAQ)

Last updated on 03/02/2026